

THE MATHEMATICS OF GENERALIZATION

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1. Mathematical framework

We observe $x \in \mathcal{X} = \{\bar{x}_1, \dots, \bar{x}_N\} \subset \mathbb{R}$, and $y \in \mathcal{Y} = \{-1, 1\}$ and we would like to predict y from x. To do so, we are given a collection of observations like x and y, i.e. our data. This allows to observe the relationship between x and y multiple times and hope to estimate it well.

Assumption 1.1. *We assume our data is drawn from a probability distribution. Therefore, we consider the sample space Ω and define the random variables*

$$X : \Omega \to \mathcal{X}, \quad \omega \mapsto x \qquad \text{and} \qquad Y : \Omega \to \mathcal{Y}, \quad \eta \mapsto y;$$
$$(X, Y) \sim p,$$

with joint law p, which is a cathegorical probability distribution over $\mathcal{X} \times \mathcal{Y} = \{\bar{x}_1, \dots, \bar{x}_N\} \times \{-1, 1\}$.

Since we want to obtain a collection of observations, we define an additional random variable.

Definition 1.2 (Dataset)**.** *Given a size n, a sample space Ω, and value spaces $\mathcal{X}, \mathcal{Y}$, a dataset D is a random variable*

$$D : \Omega^n \to \mathcal{X}^n \times \mathcal{Y}^n, \qquad D \coloneqq (X_i, Y_i)_{i=1}^n,$$
$$(\omega_1, \dots, \omega_n) \mapsto ((x_1, y_1), \dots, (x_n, y_n))$$
$$D \sim p^n$$

where the X_is and Y_is are i.i.d copies of X, Y, and X, Y, p are as in Assumption 1.1. We indicate with the lower-case d realizations of D, $d = \{(x_i, y_i)\}_{i=1}^n$. Also, we use $\mathcal{D}$ to indicate the output space of D in general, and treat $\mathcal{D} = \mathcal{X}^n \times \mathcal{Y}^n$ as a particular case.

Now, predicting y from x means we want to find the best function f such that $f(x)$ is a good approximation of y. To do so, we consider an *hypotheses space* $\mathcal{H}$, which is the set of all functions $f : \mathcal{X} \to \mathcal{Y}$ that satisfy some hypotheses we choose to make on the relationship between x and y.

Fact 1.3. *Functions in $\mathcal{H}$ are deterministic objects.*

In order to find the best predictor f, we use a *learning algorithm*, in the idea of learning our predictor from the data.

Definition 1.4 (Learning algorithm)**.** *A learning algorithm A is a deterministic function*

$$A : \mathcal{D} \to \mathcal{H}, \qquad d \mapsto \hat{f}_d$$

that given some data $d \in \mathcal{D}$, outputs a prediction function $\hat{f}_d \in \mathcal{H}$.

Remark 1.5. *The hat notation $\hat{f}$ is used only to distinguish an element of $\mathcal{H}$ that comes from a learning algorithm, or that depends on data, from a generic element of $\mathcal{H}$, indicated as f.*

Now, we are ready to define our central object, the predictor.

Definition 1.6 (Predictor)**.** *A predictor $\hat{f}_D$ is a random variable*

$$\hat{f}_D : \Omega^n \to \mathcal{H}, \qquad \hat{f}_D \coloneqq A \circ D, \qquad (\omega_i)_{i=1}^n \mapsto d \mapsto \hat{f}_d$$

that depends on some data D and on a learning algorithm A. The source of randomness is the dataset D.

Now, we would like to have a way to *evaluate* our predictors, in order to select the best one. To do so, we use a metric called *expected risk*. First, we take a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$ that takes the preidction $f(x)$ and the true value y and outputs a number corresponding to the error in the prediction.

Definition 1.7 (Expected risk)**.** *Given a predictor f and a loss function ℓ, the expected risk is a deterministic function $R : (\mathcal{X} \times \mathcal{Y}) \times \mathcal{H} \to \mathbb{R}$,*

$$R(f) \coloneqq \sum_{\bar{x} \in \mathcal{X}} \sum_{y \in \{-1,1\}} \ell(f(\bar{x}), y) p(\bar{x}, y). \tag{1}$$

The issue with using $R(f)$ as a metric, however, is that we do not know the true distribution p in general. Hence, we approximate the expected risk with a new, emprical metric, the *empirical risk*, motivated by the law of large numbers.

Definition 1.8 (Empirical risk)**.** *Given a dataset D, a predictor f, and a loss function ℓ, the empirical risk is*

$$\hat{R}_D(f) \coloneqq \frac{1}{n} \sum_{i=1}^{n} \ell(f(X_i), Y_i), \qquad D = (X_1, Y_1), \ldots, (X_n, Y_n). \tag{2}$$

It is a random variable due to the randomness in D, in the sense that $\hat{R}_D \coloneqq \hat{R} \circ D$ for a deterministic function $\hat{R} : \mathcal{H} \times \mathcal{D} \to \mathbb{R}$.

Remark 1.9. *Notice that we implicitly define the two metrics as operators applying the loss function ℓ to their inputs, but we could equivalently define them taking inputs from $\mathbb{R}$, the image space of ℓ.*

For a deterministic estimator $f \in \mathcal{H}$, the following result holds.

Proposition 1.10 (Unbiasedness of the empirical risk)**.** *Given a scalar $a \in \mathbb{R}$, the empirical risk of a is an unbiased estimator of the expected risk,*

$$\mathbb{E}[\hat{R}_D(a)] = R(a). \tag{3}$$

Proof. A proof is found in the appendix (Proposition E.4). □

2. Statistical bounds on the risk

With the technology defined above, we define an estimator given a dataset D and an hypotheses space $\mathcal{H}$ as

$$\hat{f}_D \coloneqq \arg\min_{f \in \mathcal{H}} \hat{R}_D(f). \tag{4}$$

This means that among all candidates in $\mathcal{H}$, we select the one that minimizes the empirical risk. Solving problem (4) corresponds to *fitting* the estimator to the data D or *training* the estimator on D.

Subsequently, it is natural to look for a measure of performance of $\hat{f}_D$, for instance to choose the best one among a set of different hypotheses (e.g. different architectures of a neural network), or to provide a measure of reliability of our prediction. However, our estimator is now a random variable of the data. This makes the result in Proposition 1.10 not valid anymore: the empirical risk computed on a dataset D is a biased estimate of the true, expected risk of an estimator trained on D. This intuitively comes from the correlation between datapoints, as seen in Example B.1. As a solution, classical statistical theory provides splitting the dataset into two parts $(D_{train}, D_{test}) = D$ to preserve unbiasedness. Usually, we fit the estimator to D_{train} and compute $\hat{R}$ on D_{test}, to obtain $\hat{R}_{D_{test}}$ as an unbiased estimate of the risk of $\hat{f}_{D_{train}}$.

We begin by asking the following question: how many splits of the dataset are really necessary in practice? To answer this, we want to study the bias arising from using the same dataset for training and evaluating a model.

We make the following assumptions.

Assumption 2.1 (Boundedness of loss function)**.** *Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$ and a prediction function $f : \mathcal{X} \to \mathcal{Y}$, $\exists M \in \mathbb{R}$ s.t.*

$$\ell(f(x), y) \leq M \qquad \forall x \in \mathcal{X}, y \in \mathcal{Y}.$$

Assumption 2.2 (Lipschitzness of loss function and bound at zero)**.** *Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$ and a prediction function $f : \mathcal{X} \to \mathcal{Y}$,*

(i) $\exists C > 0$ *s.t.*

$$\forall a, a' \in \mathbb{R}, |\ell(a) - \ell(a')| \leq C|a - a'|$$

(ii) $\exists C' > 0$ *s.t.*

$$|\ell(0)| \leq C'$$

Let us make an example of a loss function that satisfies these assumptions.

Example 2.3 (Logistic loss function)**.** *Let $\ell(a) = \log(1+e^{-a}), a \in \mathbb{R}$. Notice that to apply this to our setting we take the product $f(x) \cdot y$ and then apply ℓ. We have*

$$\ell'(a) = -\frac{e^{-a}}{1+e^{-a}} = -\frac{1}{1+e^{a}} \quad \Rightarrow \quad |\ell'(a)| < 1 \qquad \forall a \in \mathbb{R}$$

hence

$$C = \sup_{a \in \mathbb{R}} |\ell'| \leq 1.$$

Also,

$$\ell(0) = \log 2 = C'.$$

Proposition 2.4. *For $a \in A \subset \mathbb{R}, A \coloneqq \{a \in \mathbb{R} : |a| \leq B$ for some constant $B \in \mathbb{R}\}$, Assumption 2.2 $\Rightarrow$ Assumption 2.1.*

Proof. For $a \in A$,

$$\begin{aligned}
\ell(a) &= |\ell(a)| \\
&= |\ell(a) - \ell(0) + \ell(0)| \\
&\leq |\ell(a) - \ell(0)| + |\ell(0)| \\
&\leq C|a| + C' \\
&\leq CB + C' =: M
\end{aligned}$$

□

Notice that we can always take a *truncation map* $T : \mathbb{R} \to [-1, 1]$, defined as

$$T(a) = \begin{cases} -1 & a \leq -1 \\ a & -1 < a < 1 \\ 1 & a \geq 1, \end{cases} \tag{5}$$

so that $T \circ f(x)$ is bounded for any f and x. Hence, we consider $A = \{T \circ f(x) : f \in \mathcal{H}, x \in \mathcal{X}\}$ and $\mathcal{F} \coloneqq \{f | f : \mathcal{X} \to \mathbb{R}\}$, $\mathcal{H} \subseteq \mathcal{F}, \mathcal{H} = \{T \circ f | f \in \mathcal{F}\}$. Now, we are ready to state our main result.

Theorem 2.5 (Bounds on the bias)**.** *Consider data $D : \Omega^n \to \mathcal{X}^n \times \mathcal{Y}^n$, a predictor $\hat{f}_D \in \mathcal{H}$, and a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$. Let ℓ be as in Assumption 2.2, so that $\ell(\cdot) \leq M$ for some $M \in \mathbb{R}$ by Proposition 2.4. Then, the empirical and the expected risk of $\hat{f}_D$ satisfy the following bound*

$$\boxed{0 \leq \mathbb{E}_D[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D)] \leq 2M\sqrt{\frac{\log|\mathcal{H}|}{2n}}.} \tag{6}$$

Proof. We prove the lower bound ($0 \leq \mathbb{E}_D \ldots$) first. All the expectations are over the randomness of the data D, we drop the subscript to light notation. We have

$$\hat{R}_D(\hat{f}_D) \leq \hat{R}_D(f) \quad \forall f \in \mathcal{H}, \tag{7}$$

since $\hat{f}_D \coloneqq \arg\min_{f \in \mathcal{H}} \hat{R}_D(f)$. Then, take expectations

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq \mathbb{E}[\hat{R}_D(f)] \quad \forall f \in \mathcal{H}. \tag{8}$$

Now, since the functions in $\mathcal{H}$ are deterministic, by Proposition 1.10 we have

$$\mathbb{E}[\hat{R}_D(f)] = R(f) \quad \forall f \in \mathcal{H}, \tag{9}$$

so that

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq R(f) \quad \forall f \in \mathcal{H}. \tag{10}$$

Now take d a generic realization of D and consider in particular $f = \hat{f}_d$,

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq R(\hat{f}_d) \quad \forall d \in \mathcal{D}. \tag{11}$$

Since this is valid pointwise for any d realization of D, we have

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq R(\hat{f}_D) \quad \text{a.s.} \tag{12}$$

and, finally, taking expectations

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq \mathbb{E}[R(\hat{f}_D)] \quad \text{a.s.} \tag{13}$$

Next, we prove the upper bound ($\mathbb{E} \leq 2\sqrt{\ldots}$). For any $f \in \mathcal{H}$, $\mathbb{E}[\hat{R}_D(f)] = R(f)$. Let $Z_i \coloneqq \ell(f(x_i), y_i)$ and have

$$\hat{R}_D(f) = \frac{1}{n}\sum_i Z_i$$

We now want to apply Hoeffding's inequality (Theorem A.8 in the appendix). To do so, we need to satisfy 3 conditions:

- (i) $Z_1, \ldots, Z_n$ i.i.d.;
- (ii) $\mathbb{E}[Z_1] = \cdots = \mathbb{E}[Z_n]$;
- (iii) $a_i \leq Z_i \leq b_i$ for some $a_1, \ldots, a_n \in \mathbb{R}$ and $b_1, \ldots, b_n \in \mathbb{R}$.

We have that (i) is satisfied by $Z_i s$ being i.i.d. by construction. Condition (ii) is satisfied since all the (x_i, y_i) have the same distribution. (iii) is satisfied with $a_1 = \cdots = a_n = 0$ and $b_1 = \cdots = b_n = M$ by non-negativity of the loss function and Proposition 2.4 with our definition of $\mathcal{H}$. Now we can apply Hoeffding's. We obtain

$$\Pr[\frac{1}{n}\sum_i Z_i - \mu \geq \varepsilon] \leq e^{-2n\varepsilon^2/M^2}$$

and thus

$$\Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] \leq e^{-2n\varepsilon^2/M^2} \tag{14}$$

Now, to obtain a bound for $\hat{f}_D$ we bound it with the union of the probabilities of the events for each $f \in \mathcal{H}$. We have

$$\Pr[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D) \geq \varepsilon] \leq \bigcup_{f \in \mathcal{H}} \Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] \tag{15}$$

trivially, and also

$$\bigcup_{f \in \mathcal{H}} \Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] \leq \sum_{f \in \mathcal{H}} \Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] \leq \sum_{f \in \mathcal{H}} e^{-2n\varepsilon^2/M^2} = |\mathcal{H}| e^{-2n\varepsilon^2/M^2} \tag{16}$$

where the first inequality follows by a union bound and the second by Hoeffding. All in all, we obtain

$$\Pr[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D) \geq \varepsilon] \leq |\mathcal{H}| e^{-2n\varepsilon^2/M^2} \tag{17}$$

To obtain our initial inequality from this result on a probability, we use a technique called tail integration. The key ingredient is the following

$$\mathbb{E}[X] = \int_0^\infty \Pr(X \geq \varepsilon) \ \mathrm{d}\varepsilon \tag{18}$$

which is true for any non-negative random variable X, see Proposition A.2 in the appendix. Then, we want to use our bound (17) for the integrand. Let $\alpha \coloneqq |\mathcal{H}|$ and $c \coloneqq 2n/M^2$, so that

$$\Pr(X \geq \varepsilon) \leq \alpha e^{-c\varepsilon^2}.$$

Notice that for high values of ε, the bound is a very good one, but for $\varepsilon \gg 1$ it is bad. Indeed, we can use a better bound:

$$\Pr[X \geq \varepsilon] \leq \min\{1, \alpha e^{-c\varepsilon^2}\} \tag{19}$$

Therefore, it makes sense to split the integral in (18) in two parts with a threshold $\alpha e^{-c\varepsilon_0^2} = 1$. This means splitting the integral at $\varepsilon_0 = \sqrt{\frac{\log \alpha}{c}}$. We obtain

$$\begin{aligned}\mathbb{E}[X] = \int_0^\infty \Pr(X \geq \varepsilon) \ \mathrm{d}\varepsilon &\leq \int_0^{\sqrt{\frac{\log \alpha}{c}}} 1 \ \mathrm{d}\varepsilon + \int_{\sqrt{\frac{\log \alpha}{c}}}^\infty \alpha e^{-c\varepsilon^2} \ \mathrm{d}\varepsilon \\ &= \sqrt{\frac{\log \alpha}{c}} + \alpha \frac{1}{2\sqrt{c \log \alpha}} e^{-c\varepsilon^2} = \sqrt{\frac{\log \alpha}{c}} + \frac{1}{2\sqrt{c \log \alpha}}\end{aligned}$$

where the second integral is solved with a standard Gaussian tail bound (see Proposition A.6), and in the last equality we substituted $\alpha = 1/e^{-c\varepsilon^2}$. Finally, we can use a looser but simpler bound assuming $\alpha \geq e^{1/2}$, which is sensible since $\alpha = |\mathcal{H}|$ and $e^{1/2} < 2$,

$$\mathbb{E}[X] \leq 2\sqrt{\frac{\log \alpha}{c}} \tag{20}$$

which yields the result for $X = R(\hat{f}_D) - \hat{R}(\hat{f}_D)$, $\alpha = |\mathcal{H}|$, and $c = 2n/M^2$. □

Example 2.6 (Bound for the logistic loss function)**.** *Let us see how this bound looks like for a logistic loss function. By our definition of $\mathcal{H}$ via the truncation map, we have $B = 1$; then, $C = 1$ and $C' = \log 2$ by Example 2.3. Hence,*

$$0 \leq \mathbb{E}_D[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D)] \leq (2 + 2\log 2)\sqrt{\frac{\log |\mathcal{H}|}{2n}}. \tag{21}$$

2.1. **Rademacher Complexity and Symmetrization Trick.** In this section, we present a different bound from the one in Theorem 2.5, based on the so-called *Rademacher complexity* and on a *symmetrization trick*.

Definition 2.7 (Rademacher complexity)**.** *Given a data size n, a dataset $D = (X_i, Y_i)_{i=1}^n$, a loss function $\ell : \mathcal{Y} \to \mathcal{Y}$, and an hypotheses space $\mathcal{H} = \{f | f : \mathcal{X} \to \mathcal{Y}\}$, let ξ be a n-dimensional vector of i.i.d. Rademacher random variables (i.e. taking values 1 and -1 with equal probability). The Rademacher complexity measure $\mathfrak{R}_n$ is*

$$\mathfrak{R}_n(\mathcal{H}) \coloneqq \mathbb{E}_{\xi,D}[\sup_{f\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} \xi_i \ell(Y_i, f(X_i))] = \mathbb{E}_{\xi,D} \sup_{f\in\mathcal{H}} \langle \xi, \ell(f(X), Y)\rangle \tag{22}$$

and it is a deterministic object depending on $n, \mathcal{H}$, and the distributions of ξ and D.

Notice that to be precise we should call this quantity $\mathfrak{R}_n(\ell \circ \mathcal{H})$ (for instance following [2]), but once we fix a loss function ℓ, our quantity depends only on the elements of $\mathcal{H}$ and indeed also the supremum runs over $\mathcal{H}$.

Remark 2.8. *The Rademacher complexity measures the highest possible alignment between the function $(x, y) \mapsto \ell(f(x), y)$ and the Rademacher random variable ξ. Since the loss function is given, it is a measure of the "flexibility" of $\mathcal{H}$, hence its complexity.*

The Rademacher complexity controls the maximal deviation of the empirical risk $\hat{R}(f)$ from the true risk $R(f)$ through a symmetrization property (this is Proposition 4.2 in [1]).

Proposition 2.9 (Symmetrization)**.** *Consider data $D : \Omega^n \to \mathcal{X}^n \times \mathcal{Y}^n$ and an hypotheses set $\mathcal{H}$. Given the Rademacher complexity $\mathfrak{R}_n$ as in Definition 2.7, the empirical and the expected risk of the estimators $f \in \mathcal{H}$ satisfy the following bound*

$$\mathbb{E}_D[\sup_{f\in\mathcal{H}}(R - \hat{R}_D)(f)] \leq 2\mathfrak{R}_n(\mathcal{H}). \tag{23}$$

Proof. Let $h : (x, y) \mapsto (\ell(f(x), y))$ living in $\ell \circ \mathcal{H}$ and $Z \coloneqq (X, Y)$, so that

$$R(f) = \mathbb{E}_D[h(Z)] \qquad \text{and} \qquad \hat{R}_D(f) = \frac{1}{n}\sum_i h(Z_i).$$

The key ingredient of the proof is to introduce an independent copy of our dataset D, D', and study the random variable $h(Z') - h(Z)$, which has a symmetric distribution and hence is easier to analyze (*symmetrization trick*). We proceed as follows. Introduce $D' = (Z_i')_{i=1}^n$ identical copies of Z' and rewrite

$$\mathbb{E}_D[h(Z)] = \mathbb{E}_{D'|D}[h(Z_i')|D]. \tag{24}$$

Then,

$$\begin{aligned}
\mathbb{E}_D[\sup_{f\in\mathcal{H}}(R - \hat{R})(f)] &= \mathbb{E}_D\left[\sup_{h\in(\ell\circ\mathcal{H})} \mathbb{E}_D[h(Z)] - \frac{1}{n}\sum_i h(Z_i)\right] \\
&= \mathbb{E}_D\left[\sup_{h\in(\ell\circ\mathcal{H})} \mathbb{E}_{D'|D}\left[h(Z_i')\middle|D\right] - \frac{1}{n}\sum_i h(Z_i)\right] \\
&= \mathbb{E}_D\left[\sup_{h\in(\ell\circ\mathcal{H})} \frac{1}{n}\sum_i \mathbb{E}_{D'|D}\left[h(Z_i')\middle|D\right] - h(Z_i)\right] \\
&= \mathbb{E}_D\left[\sup_{h\in(\ell\circ\mathcal{H})} \mathbb{E}_{D'|D}\left[\frac{1}{n}\sum_i h(Z_i') - h(Z_i)\middle|D\right]\right].
\end{aligned} \tag{25}$$

Using that $f(X) \leq \sup f(X) \Rightarrow \mathbb{E}[f(X)] \leq \mathbb{E}[\sup f(X)] \Rightarrow \sup \mathbb{E}[f(X)] \leq \mathbb{E}[\sup f(X)]$ and the law of total expectations, we obtain

$$\begin{aligned}\mathbb{E}_D[\sup_{f\in\mathcal{H}}(R-\hat{R})(f)] &\leq \mathbb{E}_D\left[\mathbb{E}_{D'|D}\Big[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i h(Z_i') - h(Z_i)\Big|D\Big]\right] \\ &= \mathbb{E}_{D,D'}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i h(Z_i') - h(Z_i)\right]. \end{aligned} \tag{26}$$

Now we want to apply the symmetrization trick. Let $m \coloneqq |\mathcal{H}|$. For $i = 1, \ldots, n$, the random variables $(h_1(Z_i{}') - h_1(Z_i), \ldots, h_m(Z_i{}') - h_m(Z_i))$ are jointly symmetric. This is because Z_i and $Z_i{}'$ are identically distributed, hence

$$\big(h_j(Z_i{}') - h_j(Z_i)\big)_{j=1}^m \overset{d}{=} \big(h_j(Z_i) - h_j(Z_i')\big)_{j=1}^m. \tag{27}$$

This means that for $e_i \in \{-1, 1\}$,

$$\big(h_j(Z_i{}') - h_j(Z_i)\big)_{j=1}^m \overset{d}{=} \big(e_i(h_j(Z_i{}') - h_j(Z_i))\big)_{j=1}^m. \tag{28}$$

Using independence over the is we have

$$\left(\frac{1}{n}\sum_i h_j(Z_i{}') - h_j(Z_i)\right)_{j=1}^m \overset{d}{=} \left(\frac{1}{n}\sum_i e_i(h_j(Z_i{}') - h_j(Z_i))\right)_{j=1}^m, \tag{29}$$

and thus

$$\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i h(Z_i{}') - h(Z_i) \overset{d}{=} \sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i e_i(h(Z_i{}') - h(Z_i)), \tag{30}$$

where since the supremum is a functional of the stochastic process, equality in distribution of the supremum follows from equality in distribution of the process. Hence, we obtain

$$\mathbb{E}_{D,D'}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i h(Z_i{}') - h(Z_i)\right] = \mathbb{E}_{D,D'}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i e_i(h(Z_i{}') - h(Z_i))\right] \tag{31}$$

$$= \mathbb{E}_{D,D'|\xi}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i \xi_i(h(Z_i{}') - h(Z_i))\Big|\xi = e\right] \tag{32}$$

$$= \mathbb{E}_{D,D',\xi}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i \xi_i(h(Z_i') - h(Z_i))\right]. \tag{33}$$

where in the last row we take the expected value over the disorder of ξ and use the tower law. Plugging this into inequality (26), we obtain

$$\begin{aligned}\mathbb{E}_D[\sup_{f\in\mathcal{H}}(R-\hat{R})(f)] &\leq \mathbb{E}_{D,D',\xi}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i \xi_i(h(Z_i') - h(Z_i))\right] \\ &\leq \mathbb{E}_{D',\xi}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i \xi_i h(Z_i')\right] + \mathbb{E}_{D,\xi}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i -\xi_i h(Z_i)\right] \\ &= \mathbb{E}_{D',\xi}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i \xi_i h(Z_i')\right] + \mathbb{E}_{D,\xi}\left[\sup_{h\in(\ell\circ\mathcal{H})}\frac{1}{n}\sum_i \xi_i h(Z_i)\right] \\ &= 2\mathfrak{R}_n(\mathcal{H}) \end{aligned}$$

because $\xi h(Z) \overset{d}{=} -\xi h(Z)$. □

The following result gives us an explicit bound on the Rademacher complexity. This is Theorem 3.3 in [4].

Lemma 2.10 (Massart's lemma)**.** *Let $\ell : \mathcal{Y} \to \mathcal{Y}$ be as in Assumption 2.2, so that $\ell(\cdot) \leq M$ by Proposition 2.4. Then, the Rademacher complexity of $\mathcal{H}$ satisfies*

$$\mathfrak{R}_n(\mathcal{H}) = \mathbb{E}_{\xi,D}\left[\sup_{f\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^{n} \xi_i \ell(Y_i, f(X_i))\right] \leq M\sqrt{\frac{2\log|\mathcal{H}|}{n}}. \tag{34}$$

Proof. For any $t > 0$,

$$\begin{aligned}
\exp\left(t\,\mathbb{E}_{\xi,D}\Big[\sup_{f\in\mathcal{H}}\sum_{i=1}^{n}\xi_i\ell(Y_i,f(X_i))\Big]\right) &\leq \mathbb{E}_{\xi,D}\left[\exp\left(t\sup_{f\in\mathcal{H}}\sum_{i=1}^{n}\xi_i\ell(Y_i,f(X_i))\right)\right] \\
&= \mathbb{E}_{\xi,D}\left[\sup_{f\in\mathcal{H}}\exp\left(t\sum_{i=1}^{n}\xi_i\ell(Y_i,f(X_i))\right)\right] \\
&\leq \mathbb{E}_{\xi,D}\left[\sum_{f\in\mathcal{H}}\exp\left(t\sum_{i=1}^{n}\xi_i\ell(Y_i,f(X_i))\right)\right] \\
&= \mathbb{E}_D\left[\sum_{f\in\mathcal{H}}\prod_{i=1}^{n}\mathbb{E}_{\xi_i}\left[\exp\left(t\xi_i\ell\left(Y_i,f(X_i)\right)\right)\right]\right].
\end{aligned} \tag{35}$$

where we use Jensen's inequality on the convex function exp in the first inequality, we trivially bound the supremum with the sum, and we use the independence of the ξ_i's across $i = 1, \ldots, n$ in the last equality.

Now, we want to apply Hoeffding's lemma (Theorem A.4 in the appendix) to each factor. Notice that we can do so because each $\xi_i\ell(Y_i, f(X_i))$ is a centered random variable, since $\mathbb{E}[\xi_i] = 0$ and ξ_i is independent of $\ell(Y_i, f(X_i))$, and it satisfies $|\xi_i\ell(Y_i, f(X_i))| \leq M$ by assumption. We obtain

$$\mathbb{E}_{\xi_i}\Big[\exp\left(t\xi_i\ell(Y_i,f(X_i))\right)\Big] \leq e^{t^2M^2/2}. \tag{36}$$

Plugging this back into (35), we obtain

$$\exp\left(t\,\mathbb{E}_{\xi,D}\Big[\sup_{f\in\mathcal{H}}\sum_{i=1}^{n}\xi_i\ell(Y_i,f(X_i))\Big]\right) \leq \sum_{f\in\mathcal{H}}\prod_{i=1}^{n}e^{t^2M^2/2} = |\mathcal{H}|e^{nt^2M^2/2}, \tag{37}$$

Now take the log of both sides

$$t\,\mathbb{E}_{\xi,D}\Big[\sup_{f\in\mathcal{H}}\sum_{i=1}^{n}\xi_i\ell(Y_i,f(X_i))\Big] \leq \log|\mathcal{H}| + \frac{nt^2M^2}{2}. \tag{38}$$

and divide by $tn > 0$, obtaining

$$\mathfrak{R}_n(\mathcal{H}) \leq \frac{\log|\mathcal{H}|}{tn} + \frac{tM^2}{2}. \tag{39}$$

Since (39) holds for every $t > 0$, we look for the tightest bound. We have

$$-\frac{\log|\mathcal{H}|}{t^2n} + \frac{M^2}{2} = 0 \quad\Longrightarrow\quad t^\star = \sqrt{\frac{2\log|\mathcal{H}|}{nM^2}}, \tag{40}$$

and

$$2\frac{\log|\mathcal{H}|}{t^3n} \geq 0 \tag{41}$$

hence t^* is a minimizer. Plugging it back into (39) yields

$$\mathfrak{R}_n(\mathcal{H}) \leq M\sqrt{\frac{2\log|\mathcal{H}|}{n}}. \tag{42}$$

□

Hence, putting together Proposition 2.9 and Lemma 2.10, we obtain the following bound on the deviation of the empirical risk from the true risk

$$\boxed{\mathbb{E}_D[\sup_{f\in\mathcal{H}}(R-\hat{R}_D)(f)] \leq 2M\sqrt{\frac{2\log|\mathcal{H}|}{n}}.} \tag{43}$$

2.2. **Adding the variance.** Let us consider only deterministic estimators f for a moment (i.e. whose choice does not depend on the data, following [2], Section 5), and recall inequality (14)

$$\Pr[R(f)-\hat{R}_D(f)\geq\varepsilon] \leq e^{-2n\varepsilon^2/M^2}. \tag{44}$$

io credo loro usino una f deterministica perché così nella loro (3) il valore atteso a dx va via, e poi per poter applicare Bernstein

In this section, we want to improve this bound taking more into consideration the structure of our problem. The main ingredient to do so is Bernstein inequality (Theorem A.9 in the Appendix) that, using a Taylor expansion of the moment generating function and boundedness of the random variable, uses the second-order moment to bound tail probabilities instead of the range of the support, as Hoeffidng's inequality does. Indeed, given n random variables $\ell(f(X_i),Y_i)-\mathbb{E}_D[\ell(f(X_i),Y_i)]$, $i=1,\dots,n$, which are independent, centered, and bounded by M by Assumption 2.1 in the worst case coming from $\ell\leq M$! can do better?, $\forall\varepsilon>0$ it holds

$$\Pr[R(f)-\hat{R}_D(f)\geq\varepsilon] \leq e^{-\frac{n\varepsilon^2}{2(\mathrm{Var}(f)+M\varepsilon/3)}}. \tag{45}$$

come posso giustificare che metto prima l'exp value e poi la media empirica? si ricollega al dubbio su Hoeffding nella A.8 Notice that with $\mathrm{Var}(f)$ we mean $\mathrm{Var}(\ell(f(X),Y))$, since the fs are deterministic. If $\mathrm{Var}(f)\leq\varepsilon$, i.e. if the expected risk is small, the upper bound behvaes like $e^{n\varepsilon}$, improving on (44). We show this in Remark D.1 in the Appendix, where we also obtain the high-probability version of (45), stated in [2].

Ora, quella detta sopra è solo la logica generale di questa sezione. I risultati concreti (che portano avanti il bound) invece partono ora e boundano il sup della deviazione, non la deviazione per una f singola come la (44) e (45). Questi risultati sono il teorema 5.1, il quale è ottenuto con simmetrizzazione e Rademacher. Qui non mi è chiaro perché sarebbe un risultato sulla varianza: forse per la normalizzazione, ma questa è presentata come un fatto tecnico e Bernstein dovrebbe essere il tool da usare per includere la varianza, ma qui non lo usano.

Poi, si applica questo scenario (con una diseguaglianza algebrica scema) al caso di risk minimization, e si ottiene il corollario 5.2, e poi si fa un ragionamento X che porta a fornire una versione del bound sopra con $\inf R(f)$, il corollario 5.3, ottenuto usando Bernstein + 5.2. Non mi è chiaro nemmeno questo ragionamento X. Riassumendo, quello che ho capito è

$$\sup_{f\in\mathcal{H}}\frac{Pf-P_nf}{\sqrt{Pf}} \xrightarrow{\text{Symm. e Rademacher}} \text{Thm 5.1} \xrightarrow{\text{Apply to risk min. + algebraic ineq.}} \text{Cor 5.2} \xrightarrow{\text{Ragionamento } X} \text{Cor 5.3}$$

e le mie domande sono perché il thm 5.1 ingloba la varianza e il ragionamento X.

Qui scrivo i risultati della 5.2 e 5.3 nel mio setting, cioè invece della VC dimension uso la cardinalità di $\mathcal{H}$ e scrivo le diseguaglianze in forma di ε e non δ.

We obtain the following bounds:

$$\boxed{\Pr[(R-\hat{R}_D)(\hat{f}_D)>\varepsilon] \leq |\mathcal{H}|\exp\left(-\frac{n}{16}\left(\sqrt{\hat{R}_D(\hat{f}_D)+4\varepsilon}-\sqrt{\hat{R}_D(\hat{f}_D)}\right)^2\right).} \tag{46}$$

and

$$\boxed{\Pr[R(\hat{f}_D) - \inf_{f\in\mathcal{H}} R(f) > \varepsilon] \leq |\mathcal{H}| \exp\left(-\frac{n}{4}\left(\sqrt{\inf_{f\in\mathcal{H}} R(f) + 4\varepsilon} - \sqrt{\inf_{f\in\mathcal{H}} R(f)}\right)^2\right)} \tag{47}$$

where we arbitrarily set the constant C to 1. The first bound simplifies when the training error goes to 0 and the second bound when there is an f that achieves zero test error. I.e., in the former we need a long enough training or enough parameters, in the latter we need the right model (possibly, one with few parameters) to be included in the hypotheses class.

Noto che nella (47) la varianza c'è anche se nascosta, perché il ragionamento che fanno nel paper è il seguente.

Let $f' \in \mathcal{H}$ be such that $R(f') = \inf_{f\in\mathcal{H}} R(f)$. We have

$$\hat{R}_D(\hat{f}_D) \leq \hat{R}_D(f')$$

by definition of $\hat{f}_D$. Then,

$$\hat{R}_D(\hat{f}_D) - R(f') \leq \hat{R}_D(f') - R(f')$$

applying Bernstein inequality,

$$\hat{R}_D(\hat{f}_D) - R(f') \leq \sqrt{\frac{2R(f')\log\frac{1}{\delta}}{n}} + \frac{2\log\frac{1}{\delta}}{3n}.$$

Then we say that $R(\hat{f}_D)$ is bounded by $\hat{R}_D(\hat{f}_D)$ as in (46), and hence we can bound $R(\hat{f}_D) - R(f')$ using (46) and the last inequality.

This would hold for any function $f \in \mathcal{H}$, ma usiamo la varianza della funzione che minimizza il rischio atteso (valore atteso), e la usiamo in $\mathrm{Var}(f') \leq R(f') = \inf_{f\in cH} R(f)$. Questo permette di studiare il caso in cui questo infimo è zero, cioè quando lo stimatore giusto è incluso nella classe di ipotesi. E questo ritorna utile nella sezione dopo, con l'errore.

Finally, we use tail integration to obtain a result in expected value from inequality (45), just like we did to go from inequality (17) to (20), our first bound (although it's slightly more complixated here).

First, recall the relationship between the expected value of a non-negative random variable and a tail probability bound,

$$\mathbb{E}[X] = \int_0^\infty \Pr[X \geq \varepsilon] \ \mathrm{d}\varepsilon.$$

2.3. **Noise analysis.** We report the results given by section 5.3 of [2] and specify them to our setting.

Let $\ell : \mathcal{Y} \times \mathcal{Y} \to \{0, 1\}$ be the counting loss, $\ell(f(X), Y) = \mathbf{1}_{\{f(X)\neq Y\}}$ and consider a new set of functions $\mathcal{G} \coloneqq \left\{\mathbf{1}_{\{f(X)\neq Y\}} - \mathbf{1}_{\{f_{\mathrm{opt}}(X)\neq Y\}} | f \in \mathcal{H}\right\}$ where $f_{\mathrm{opt}} \in \arg\min_{f\in\mathcal{H}} R(f)$ (this is indicated by $\mathcal{F}$ in the paper). We will indicate the elements of $\mathcal{G}$ as g_f for each $f \in \mathcal{H}$. Then, $\mathbb{E}[g_f] = \mathbb{E}[\ell(f)] - \mathbb{E}[\ell(f_{\mathrm{opt}})] = R(f) - R(f_{\mathrm{opt}})$.

Theorem 2.11 (Bound with noise)**.** *Consider the setting above. Assuming that*

$$\mathrm{Var}(f - f_{\mathrm{opt}}) \leq \left(\frac{R(f) - R(f_{\mathrm{opt}})}{h}\right)^\alpha \tag{48}$$

for some $h > 0$, $\alpha \in (0, 1]$, we have

$$R(\hat{f}_D) - \inf_{f\in\mathcal{H}} R(f) \leq \left(2\frac{\log\frac{|\mathcal{H}|}{\delta}}{nh^\alpha}\right)^{\frac{1}{2-\alpha}} + \frac{8\log\frac{|\mathcal{H}|}{\delta}}{3n} \tag{49}$$

with probability at least $1 - \delta$.

Proof. First, we apply Bernstein's inequality (45) plus a union bound over $\mathcal{G}$ to the random variables $g_f(f(X_i), Y_i)$. We obtain that with probability at least $1-\delta$, $\forall g \in \mathcal{G}$ it holds

$$R(f) - R(f_{\text{opt}}) - \left(\hat{R}_D(f) - \hat{R}_D(f_{\text{opt}})\right) \leq \sqrt{\frac{2((R(f) - R(f_{\text{opt}}))/h)^{\alpha} \log \frac{|\mathcal{H}|}{\delta}}{n}} + \frac{4 \log \frac{|\mathcal{H}|}{\delta}}{3n} \tag{50}$$

where we assume that $f \mapsto g_f$ is bijective.[1] Then, we apply this bound to $f = \hat{f}_D$. Using that $\hat{R}_D(\hat{f}_D) - \hat{R}_D(f_{\text{opt}}) \leq 0$, we obtain that with probability at least $1-\delta$,

$$R(\hat{f}_D) - R(f_{\text{opt}}) \leq \sqrt{\frac{2\left(\left(R(\hat{f}_D) - R(f_{\text{opt}})\right)/h\right)^{\alpha} \log \frac{|\mathcal{H}|}{\delta}}{n}} + \frac{4 \log \frac{|\mathcal{H}|}{\delta}}{3n}. \tag{51}$$

Now let $Z \coloneqq R(\hat{f}_D) - R(f_{\text{opt}}) \geq 0$ and $l \coloneqq \log \frac{|\mathcal{H}|}{\delta}$, obtaining

$$Z \leq \sqrt{\frac{2Z^{\alpha} l}{nh^{\alpha}}} + \frac{4l}{3n} = Z^{\alpha/2} A + B \tag{52}$$

where $A = \sqrt{\frac{2l}{nh^{\alpha}}}$ and $B = \frac{4l}{3n}$. We want to apply Young's inequality (Lemma A.10 in the Appendix) to $x = Z^{\alpha/2}, y = A, p = \frac{2}{\alpha}, q = \frac{2}{2-\alpha}$. We obtain that

$$Z^{\alpha/2} A \leq \frac{\alpha}{2} Z + \frac{2-\alpha}{2} A^{\frac{2}{2-\alpha}} \tag{53}$$

since $x^p = Z$ and $y^q = A^{\frac{2}{2-\alpha}}$. Plugging it in,

$$Z \leq \frac{\alpha}{2} Z + \frac{2-\alpha}{2} A^{\frac{2}{2-\alpha}} + B \quad \iff \quad \frac{2-\alpha}{2} Z \leq \frac{2-\alpha}{2} A^{\frac{2}{2-\alpha}} + B \tag{54}$$

and dividing by $\frac{2-\alpha}{2} > 0$ we get

$$Z \leq A^{\frac{2}{2-\alpha}} + \frac{2B}{2-\alpha} = \left(\frac{2l}{nh^{\alpha}}\right)^{\frac{1}{2-\alpha}} + \frac{8l}{3n(2-\alpha)}. \tag{55}$$

Finally, $\alpha \in (0, 1]$ implies $2 - \alpha \geq 1$, so the last term is at most $\frac{8l}{3n}$. Hence, with probability at least $1-\delta$,

$$R(\hat{f}_D) - R(f_{\text{opt}}) \leq \left(\frac{2 \log \frac{|\mathcal{H}|}{\delta}}{nh^{\alpha}}\right)^{\frac{1}{2-\alpha}} + \frac{8 \log \frac{|\mathcal{H}|}{\delta}}{3n}. \tag{56}$$

The second term is of order $\frac{l}{n}$, which is of lower order than the first one when $\alpha < 1$ (as $\frac{1}{2-\alpha} < 1$), while for $\alpha = 1$ they are of the same order and the bound is $O\left(\frac{l}{n}\right)$: the fast rate. □

It remains to discuss the assumption that $\text{Var}(f - f_{\text{opt}}) \leq \left(\frac{R(f) - R(f_{\text{opt}})}{h}\right)^{\alpha}$.

The equation from the paper, (18), gives the following bound:

$$\boxed{\Pr\left[R(\hat{f}_D) - \inf_{f \in \mathcal{H}} R(f) \geq \varepsilon\right] \leq |\mathcal{H}| \exp\left(-\frac{n}{2} h^{\alpha} \varepsilon^{2-\alpha}\right).} \tag{57}$$

we try and implement this, and I will discuss it if it's useful.

Let's consider a simple case to grasp an intuition. If we assume that the best predictor (the Bayes classifier, [2]) is included in $\mathcal{H}$, then the constant h is such that the a posteriori probability is

$$|2 \Pr[Y | X = \bar{x}] - 1| > h \qquad \forall \bar{x} \in \mathcal{X}.$$

[1] $\star$ this assumption is a great source of improvements, as instead of $|\mathcal{H}|$ the bound presents more in general $|\mathcal{G}|$ which might be smaller than the hypotheses set.

This means that h controls how distant the a posteriori probability is from $1/2$: if h is small, then this probability may be close to $1/2$ which means we have a large noise in the data, if h is large we have less noise. This whole reasoning is valid when $\alpha = 1$. More generally, the correct interpretation of these two values is given by the assumption on the variance.

2.4. **Refined Massart bound.** Notation fix: Let us use h to indicate functions in $\mathcal{H}$ and $\ell_h : \mathcal{X} \times \mathcal{Y} \to \{0, 1\}$ to indicate functions in $\mathcal{L} \coloneqq \ell \circ \mathcal{H}$, with $\ell_h(x, y) = \ell(h(x), y)$.

Recall

Definition 2.12 (Rademacher complexity)**.**

$$\mathcal{R}_n(\mathcal{L}) = \mathbb{E}_{\xi, D}\left[\sup_{\ell_h \in \mathcal{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell(h(X_i), Y_i)\right]$$

Theorem 2.13 (Rademacher complexity bound)**.**

$$\mathbb{E}_D[(R - \hat{R})(\hat{h})] \leq 2\mathcal{R}_n(\mathcal{L})$$

Where the bound holds for the gap measured at $\hat{h}$ because it holds for the supremum of the gap over $h \in \mathcal{H}$ (Proposition 2.9), and hence for every function in $\mathcal{H}$.

Then, the application of Massart lemma (Lemma 2.10) can be refined by counting functions in $\mathcal{L}$ instead of $\mathcal{H}$.

Lemma 2.14 (Massart)**.**

$$\mathcal{R}_n(\mathcal{L}) \leq \mathbb{E}_D\left[M\sqrt{\frac{2 \log N_D(\mathcal{H})}{n}}\right]$$

where $N_D(\mathcal{H}) = |\{(\ell_h(X_1, Y_1), \ldots, \ell_h(X_n, Y_n)) : h \in \mathcal{H}\}|$.

Proof. This is Lemma 2.10 with the observation that we can take the supremum over loss vectors $\ell_h(X_i, Y_i)$ instead of over functions h. □

Lemma 2.15 (Counting inequalities)**.**

$$N_D(\mathcal{H}) \leq \min\{|\mathcal{H}|, 2^n\}.$$

Proof. Given a dataset, its realizations through ℓ_h are less than or equal to those through h because applying ℓ cannot increase the number of outputs, and the number of images of the functions in $\mathcal{H}$ are less than or equal to the number of functions $|\mathcal{H}|$. On the other hand, $N_D(\mathcal{H}) \leq 2^n$ because we have at most 2^n n-long sequences of 0s and 1s. □

Combining Theorem 2.13, Lemma 2.14 and Lemma 2.15, we obtain

$$\mathbb{E}_D[(R - \hat{R})(\hat{h})] \leq 2M\sqrt{\frac{2 \min\{\log |\mathcal{H}|, \log 2^n\}}{n}} \leq \min\left\{O\left(\sqrt{\frac{\log |\mathcal{H}|}{n}}\right), O(1)\right\}. \tag{58}$$

2.5. **Dependence on the best risk.** Set

$$R^*_{\mathcal{H}} = \min_{h \in \mathcal{H}} R(h).$$

We aim at obtaining a bound on the gap at $\hat{h}$ in terms of the best risk $R^*_{\mathcal{H}}$.

Theorem 2.16 (Bound with empirical risk)**.** *With probability at least* $1 - \delta$*,*

$$R(\hat{h}) - \hat{R}(\hat{h}) \leq \sqrt{\frac{\hat{R}(\hat{h}) \log(2N_{2n}(\mathcal{H})/\delta)}{n}} + 16\frac{\log(2N_{2n}(\mathcal{H})/\delta)}{n}$$

where $N_n(\mathcal{H}) \coloneqq \sup_{(x_1, y_1), \ldots, (x_n, y_n) \in \mathcal{X} \times \mathcal{Y}} N_D(\mathcal{H})$.

Proof. Apply the relative deviation inequality,

$$\Pr\left[\exists h \in \mathcal{H} : R(h) - \hat{R}(h) > \varepsilon\sqrt{R(h)}\right] \leq 2N_{2n}(\mathcal{H})e^{-2\varepsilon^2/16}$$

with the suitable ε and then isolate $\hat{R}(\hat{h})$ on the RHS. □

Corollary 2.17 (Bound with best risk)**.** *With probability at least* $1-\delta$*,*

$$R(\hat{h}) - \hat{R}(\hat{h}) \leq \sqrt{\frac{R^*_{\mathcal{H}} \log(4N_{2n}(\mathcal{H})/\delta)}{n}} + 20\frac{\log(4N_{2n}(\mathcal{H})/\delta)}{n}$$

Proof. Set $h_0 = \arg\min_{h\in\mathcal{H}} R(h)$, so that $R(h_0) = R^*_{\mathcal{H}}$. Then,

$$\hat{R}(\hat{h}) \leq \hat{R}(h_0). \tag{59}$$

Since h_0 is independent of the data, $\mathbb{E}_D[\hat{R}(h_0)] = R(h_0)$ and hence Bernstein gives, with probability at least $1-\delta/2$,

$$\hat{R}(h_0) \leq R^*_{\mathcal{H}} + \sqrt{\frac{2R *_{\mathcal{H}} \log(2/\delta)}{n}} + \frac{\log(2/\delta)}{3n}.$$

Since $N_{2n}(\mathcal{H}) \geq 1$ and thus $\frac{\log(2/\delta)}{n} \leq \frac{\log(4N_{2n}(\mathcal{H})/\delta)}{n}$,

$$\hat{R}(h_0) \leq \left(\sqrt{R^*_{\mathcal{H}}} + \sqrt{\frac{4\log(N_{2n}(\mathcal{H})/\delta)}{n}}\right)^2 \tag{60}$$

Substitute (59) into (60), and apply the bound to $\hat{R}(\hat{h})$ in the RHS of Theorem 2.16 at confidence $1-\delta/2$. This yields the desired bound, which holds with probability at least $1-\delta$ by a union bound on the events ((60) and Theorem 2.16) which we require to hold simultaneously. □

Definition 2.18 (Bayes risk)**.**

$$R^* \coloneqq \inf_{h:\mathcal{X}\to\mathcal{Y}} R(h) = \sigma^2$$

Definition 2.19 (Approximation error)**.**

$$A_{\mathcal{H}} = R^*_{\mathcal{H}} - R^*$$

Corollary 2.20 (Noise and approximation error bound)**.** *With probability* $1-\delta$*,*

$$R(\hat{h}) - \hat{R}(\hat{h}) \leq \sqrt{\frac{(A_{\mathcal{H}} + \sigma^2)\log(4N_{2n}(\mathcal{H})/\delta)}{n}} + 20\frac{\log(4N_{2n}(\mathcal{H})/\delta)}{n} \tag{61}$$

Proof. Substitute $R^*_{\mathcal{H}} = A_{\mathcal{H}} + \sigma^2$ into Corollary 2.17. □

Where

- σ^2 measures the irreducible classification error due to noise;
- $A_{\mathcal{H}}$ measures the approximation error of the hypotheses class.

Notice that when $R^*_{\mathcal{H}} = 0$, the bound becomes linear in $\log(N_{2n}(\mathcal{H}))$.

Corollary 2.21 (Noise and approximation error bound in expectation)**.**

$$\mathbb{E}_D\left[R(\hat{h}) - \hat{R}(\hat{h})\right] \leq \sqrt{\frac{(A_{\mathcal{H}} + \sigma^2)\left(\log(4N_{2n}(\mathcal{H})) + 1\right)}{n}} + 20\,\frac{\log(4N_{2n}(\mathcal{H})) + 1}{n}$$

Proof. Let $\text{gap} = (R - \hat{R}(\hat{h}))$ and $C = \sqrt{\frac{\log(4N_n(\mathcal{H})/\delta)}{n}}$. Corollary 2.17 gives

$$\Pr\left[\text{gap} \le \sqrt{R^*_{\mathcal{H}}}\, C + 20C^2\right] \ge 1 - \delta$$
$$\implies \Pr\left[\text{gap} > \sqrt{R^*_{\mathcal{H}}}\, C + 20C^2\right] \le \delta$$

Set $\varepsilon = \sqrt{R^*_{\mathcal{H}}}\, C + 20C^2$. Let us now find $\delta(\varepsilon)$:

$$20C^2 + \sqrt{R^*_{\mathcal{H}}}\, C - \varepsilon = 0 \implies C = \frac{-\sqrt{R^*_{\mathcal{H}}} + \sqrt{R^*_{\mathcal{H}} + 80\varepsilon}}{40}$$

So we have

$$\frac{\log(4N_{2n}(\mathcal{H})/\delta)}{n} = \left(\frac{\sqrt{R^*_{\mathcal{H}} + 80\varepsilon} - \sqrt{R^*_{\mathcal{H}}}}{40}\right)^2$$
$$\frac{4N_{2n}(\mathcal{H})}{\delta} = \exp\left(n\left(\frac{\sqrt{R^*_{\mathcal{H}} + 80\varepsilon} - \sqrt{R^*_{\mathcal{H}}}}{40}\right)^2\right)$$

and hence

$$\delta(\varepsilon) = 4N_{2n}(\mathcal{H}) \exp\left(-\frac{n}{1600}\left(\sqrt{R^*_{\mathcal{H}} + 80\varepsilon} - \sqrt{R^*_{\mathcal{H}}}\right)^2\right).$$

Using that $\mathbb{E}[X] = \int_0^\infty \Pr[X \ge \varepsilon]\, d\varepsilon$ for $X \ge 0$, applied to $\text{gap}^+ \coloneqq \max\{\text{gap}, 0\}$,

$$\mathbb{E}_D[\text{gap}] \le \mathbb{E}_D[\text{gap}^+] = \int_0^\infty \Pr[\text{gap} \ge \varepsilon]\, d\varepsilon \le \int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon.$$

By a shift of perspective, looking at the integral as a sum of horizontal stripes instead of vertical stripes in the graph of $\varepsilon \mapsto \delta(\varepsilon)$, we obtain

$$\int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon = \int_0^1 \delta^{-1}(t)\, dt = \int_0^1 \varepsilon(\delta)\, d\delta$$

Hence, writing $\varepsilon(\delta) = \sqrt{R^*_{\mathcal{H}}}\, C + 20C^2$ explicitly,

$$\mathbb{E}_D[\text{gap}] \le \int_0^1 \varepsilon(\delta)\, d\delta = \sqrt{\frac{R^*_{\mathcal{H}}}{n}} \int_0^1 \sqrt{\log \frac{4N_{2n}(\mathcal{H})}{\delta}}\, d\delta + \frac{20}{n} \int_0^1 \log \frac{4N_{2n}(\mathcal{H})}{\delta}\, d\delta.$$

The logarithm is the only part depending on δ, and its integral is

$$\int_0^1 \log \frac{4N_{2n}(\mathcal{H})}{\delta}\, d\delta = \log(4N_{2n}(\mathcal{H})) - \int_0^1 \log \delta\, d\delta = \log(4N_{2n}(\mathcal{H})) + 1.$$

Applying Jensen's inequality to the square root term,

$$\int_0^1 \sqrt{\log \frac{4N_{2n}(\mathcal{H})}{\delta}}\, d\delta \le \sqrt{\int_0^1 \log \frac{4N_{2n}(\mathcal{H})}{\delta}\, d\delta} = \sqrt{\log(4N_{2n}(\mathcal{H})) + 1},$$

we obtain

$$\mathbb{E}_D[\text{gap}] \le \sqrt{\frac{R^*_{\mathcal{H}}\left(\log(4N_{2n}(\mathcal{H})) + 1\right)}{n}} + 20\,\frac{\log(4N_{2n}(\mathcal{H})) + 1}{n},$$

which is the claim since $R^*_{\mathcal{H}} = A_{\mathcal{H}} + \sigma^2$.

□

Asymptotically,

$$\mathbb{E}_D\left[R(\hat{h}) - \hat{R}(\hat{h})\right] \leq O\left(\sqrt{\frac{(A_{\mathcal{H}} + \sigma^2)\,\log N_{2n}(\mathcal{H})}{n}} + \frac{\log N_{2n}(\mathcal{H})}{n}\right)$$

because, at the numerator,

$$\log(4N_{2n}(\mathcal{H})) + 1 = \log 4 + \log N_{2n}(\mathcal{H}) + 1 = \log N_{2n}(\mathcal{H}) + O(1)$$

and for large x,

$$\log x + O(1) \leq C \log x$$

for some constant $C > 0$. Plugging in the counting inequality from Lemma 2.15 with $2n$, we obtain and plot the following bound

$$\mathbb{E}_D\left[R(\hat{h}) - \hat{R}(\hat{h})\right] \leq \min\left\{O\left(\sqrt{\frac{(A_{\mathcal{H}} + \sigma^2)\log|\mathcal{H}|}{n}} + \frac{\log|\mathcal{H}|}{n}\right), O\left(\sqrt{A_{\mathcal{H}} + \sigma^2} + 1\right)\right\}$$

Remark: recall that $N_{2n}(\mathcal{H})$ is the maximum number of loss vectors over the union of two datasets D, D', or equivalently over a dataset with size $2n$. The counting lemma must be applied to $2n$-long loss vectors. Hence, we still have $N_{2n}(\mathcal{H}) \leq |\mathcal{H}|$ because the number of functions is still the same, and the bound counting all the possible sequences becomes $N_{2n}(\mathcal{H}) \leq 2^{2n}$.

The experimental design is the same as before, except for the fact that we use a Bernoulli error with $p = 0.8$, i.e.

$$y = \text{sign}(h^*(x)) \cdot b, \qquad b \sim \text{Bernoulli}(p)$$

This eases the computation of the Bayes risk, which is $R^* = p$ since $p \geq 0.5$. The minimum risk over our hypotheses set $R^*_{\mathcal{H}}$, instead, is computed using dynamic programming. We produce labels using our sine function and the Bernoulli error and we cluster the intervals where labels have the same sign, obtaining M contiguous blocks. We create a dp $k \times 2$ matrix that stores the minimum risk for each number of maximum splits k, up to $\min(M-1, 2^d)$ where d is the maximum depth we are considering, when the last assigned label is 1 or -1 (the columns). Then, for each contiguous block we dynamically update the dp matrix summing the minimum risk among all trees with at most k splits. In the end, we select the minimum risk among the tree predicting 1 or -1 in the very last block.

2.6. **Growth function.**

Definition 2.22 (Growth function)**.** *The growth function of $\mathcal{H}$ is*

$$\Pi_n(\mathcal{H}) \coloneqq \sup_{x_1,\ldots,x_n \in \mathcal{X}} \left|\{(h(x_1), \ldots, h(x_n)) : h \in \mathcal{H}\}\right|.$$

The following extends Lemma 2.15.

Lemma 2.23 (Counting inequalities with growth function)**.** *For every sample D of size n,*

$$1 \leq N_D(\mathcal{H}) \leq \Pi_n(\mathcal{H}) \leq \min\{|\mathcal{H}|, 2^n\}.$$

Proof. $N_D(\mathcal{H}) \leq \Pi_n(\mathcal{H})$ because the number of distinct loss vector can at most decrease, given prediction vectors. $\Pi_n(\mathcal{H}) \leq |\mathcal{H}|$ because the realizations of functions in h are at most as much as the functions in h (equality when functions in h are bijective). $\Pi_n(\mathcal{H}) \leq 2^n$ because we have at most 2^n sequences of n 1s and -1s. □

Corollary 2.24 (Growth-function bound)**.** [1, Section 3.2, Corollary 3.8]

$$\mathcal{R}_n(\mathcal{L}) \leq \mathbb{E}_D\left[M\sqrt{\frac{2\log N_D(\mathcal{H})}{n}}\right] \leq M\sqrt{\frac{2\log \Pi_n(\mathcal{H})}{n}}. \tag{62}$$

Consequently,

$$\mathbb{E}_D[(R-\hat{R})(\hat{h})] \leq 2M\sqrt{\frac{2\log \Pi_n(\mathcal{H})}{n}}. \tag{63}$$

Proof. Inequality (62) is Lemma 2.14 combined with Lemma 2.23. Inequality (63) follows similarly, from Theorem 2.13 and (62). □

Applying Lemma 2.23 to (63) yields the saturating bound (58).

References

[1] Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar. *Foundations of Machine Learning*. Second edition, MIT Press, 2018. `https://cs.nyu.edu/~mohri/mlbook/`.

[2] Peter L. Bartlett and Gábor Lugosi. An inequality for uniform deviations of sample averages from their means. *Statistics & Probability Letters*, 44(1):55–62, 1999. doi:10.1016/S0167-7152(98)00291-0. Author manuscript: `https://econ-papers.upf.edu/papers/280.pdf`.

2.7. **Localized Rademacher complexity.** Until now, we have seen that we have a bound

$$\min\left\{\mathcal{R}_n(\mathcal{H}_d), O\left(\sqrt{\frac{d}{n}}\right)\right\}$$

that saturates, but is not sensitive to the noise, the ratio n/N, or the complexity of the data-generating function. Then, in the last two sections we saw bounds capturing the variance of the predictor or the noise, but which do not saturate, instead they grow linearly in the depth d. In this section, we look for a bound that saturates and is aware of the structure of the learning problem.

The motivating idea comes from the observation that the Rademacher complexity measures the expressiveness of the whole hypotheses set $\mathcal{H}$, as the supremum runs over all the functions in this set. However, we know that while learning, we will focus on a subset of the functions in $\mathcal{H}$ that are close to the true function f^*, and this happens after a few splits. Therefore, it makes sense to try to consider a *local* measure of complexity.

Hence, we define a localized loss class for a radius $r > 0$ as

$$(\ell \circ \mathcal{H}_d)_n(r) \coloneqq \left\{ f \in \mathcal{H}_d | \mathbb{E}_D\left[\frac{1}{n}\sum_i \ell(f(X_i), Y_i) \leq r\right]\right\} \tag{64}$$

Definition 2.25 (Localized Rademacher complexity)**.** *Given our usual setting and a radius $r > 0$, the localized Rademacher complexity is*

$$\mathcal{R}_n(\mathcal{H}_d(r)) \coloneqq \mathbb{E}_{\xi,D}\left[\sup_{f\in\mathcal{H}_d(r)} \frac{1}{n}\sum_i \xi_i \ell(f(X_i), Y_i)\right] \tag{65}$$

which is just a Rademacher complexity where the sup runs over all the function with expected average loss in the ball of radius r.

Then, the following result holds.

Theorem 2.26 (Fixed Point Theorem)**.**

Pensieri notturni. Let

$$D = ((X_1, Y_1), \dots, (X_n, Y_n))$$

be the training sample. In the noiseless setting,

$$Y_i = t(X_i), \qquad t : \mathcal{X} \to \{0, 1\}.$$

Let $\mathcal{H}_d$ denote the class of one-dimensional decision trees of depth at most d, and let

$$\ell : \{0, 1\} \times \{0, 1\} \to \{0, 1\}, \qquad \ell(\hat{y}, y) = \mathbf{1}\{\hat{y} \neq y\},$$

be the misclassification loss.

The empirical Rademacher complexity of the loss class is

$$\widehat{\mathfrak{R}}_D(\ell \circ \mathcal{H}_d) = \frac{1}{n} \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}_d} \sum_{i=1}^{n} \sigma_i \, \ell(h(X_i), Y_i) \right].$$

Since $Y_i = t(X_i)$, this can be written as

$$\widehat{\mathfrak{R}}_D(\ell \circ \mathcal{H}_d) = \frac{1}{n} \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}_d} \sum_{i=1}^{n} \sigma_i \, \ell(h(X_i), t(X_i)) \right].$$

The empirical localized loss class is

$$(\ell \circ \mathcal{H}_d)_D(r) = \left\{ (x, y) \mapsto \ell(h(x), y) : h \in \mathcal{H}_d, \ \frac{1}{n} \sum_{i=1}^{n} \ell(h(X_i), Y_i) \leq r \right\}.$$

hence we can compute the empirical Rademacher complexity of the empirical localized loss class, which is a measure of complexity only for a subset of the potential predictors $h \in \mathcal{H}_d$.

References

[1] Francis Bach. *Learning theory from first principles*. MIT press, 2024.

[2] Stéphane Boucheron, Olivier Bousquet, and Gábor Lugosi. Theory of classification: A survey of some recent advances. *ESAIM: probability and statistics*, 9:323–375, 2005.

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Appendix A. Mathematical tools (concentration inequalities basics)

Proposition A.1 (Linear, continuous functional commute with the Bochner integral)**.** *Given a function $f : V \to \mathcal{W}$, $V, W \subseteq \mathbb{R}^d$ and a continuous functional $\ell : V \to \mathbb{R}$, we have*

$$\ell\left(\int f(x)\ \mathrm{d}x\right) = \int \ell \circ f\ \mathrm{d}x \tag{66}$$

and in particular, when $\ell = \langle \cdot, \cdot \rangle$,

$$\left\langle v, \int f(x)\ \mathrm{d}x \right\rangle = \int \langle v, f(x) \rangle\ \mathrm{d}x \tag{67}$$

where the result holds also assuming boundedness instead of continuity.

Proof. We will prove the result for the specific case of the inner product in the finite dimensional case. The infinite dimensional case follows proving the result for simple functions, and then by an approximation algorithm. □

Proposition A.2 (Alternative characterization of expected value)**.** *For any non-negative random variable Z, it holds*

$$\mathbb{E}[Z] = \int_0^\infty \Pr[Z \geq \varepsilon]\ \mathrm{d}\varepsilon. \tag{68}$$

Proof. The proof follows directly from the definition of expected value and the Fubini-Tonelli theorem. First observe that, for any $z \in \mathbb{R}^+$,

$$z = \int_0^z \mathrm{d}\varepsilon = \int \mathbb{I}(0 \leq \varepsilon \leq z)\ \mathrm{d}\varepsilon. \tag{69}$$

Hence

$$\begin{aligned}
\mathbb{E}[Z] &= \int_0^\infty z p(z)\ \mathrm{d}z \\
&= \int_0^\infty \int \mathbb{I}(0 \leq \varepsilon \leq z)\ \mathrm{d}\varepsilon\ p(z)\ \mathrm{d}z \\
&= \int \int_0^\infty \mathbb{I}(0 \leq \varepsilon \leq z) p(z)\ \mathrm{d}z\ \mathrm{d}\varepsilon \\
&= \int \Pr[Z \geq \varepsilon]\ \mathrm{d}\varepsilon
\end{aligned}$$

where in the third line we used Fubini-Tonelli, since the integrand is non-negative, and in the last line we observe that the expected value of an indicator function is the probability that the event is true. □

Intuition. This formula is passing through the x-axis (with values of the support) and summing the probability that X is greater than x each time (weighted by the infinitesimal width of the interval $\mathrm{d}x$). Consider first that if our sample contains a lot of realizations of a large value of X, this value will contribute to each addend with a lot of probability, hence yielding a larger result (the expected value of X). If instead the opposite is true, a lot of probability will be used up by the first addends, obtaining a smaller result. Then, there's an explanation that goes like dividing the area into horizontal stripes instead of vertical, but it is not clear to me.

Theorem A.3 (Markov inequality)**.** *Given a non-negative random variable X and $t > 0$, it holds*

$$\Pr[X \geq t] \leq \frac{\mathbb{E}[X]}{t}. \tag{70}$$

Proof. The proof is a simple derivation using the definition of expected value, namely

$$\begin{aligned}
\mathbb{E}[X] &= \int_0^\infty xp(x)\ \mathrm{d}x && (71)\\
&\geq \int_0^t xp(x)\ \mathrm{d}x && (72)\\
&\geq \int_0^t tp(x)\ \mathrm{d}x && (73)\\
&= t\Pr[X \geq t]. && (74)
\end{aligned}$$

The first step is just that $\mathbb{E}[X] \geq \mathbb{E}[X|X \leq t]$ for any $t \geq 0$, and the second is algebraic manipulation. An alternative proof uses the tail integral representation of the expected value, Proposition A.2 (since X is non-negative by assumption):

$$\begin{aligned}
\mathbb{E}[X] &= \int_0^\infty \Pr[X \geq t]\ \mathrm{d}t && (75)\\
&\geq \int_0^s \Pr[X \geq t]\ \mathrm{d}t && (76)\\
&\geq \int_0^s \Pr[X \geq s]\ \mathrm{d}t && (77)\\
&= \Big[t\Pr[X \geq s]\Big]_0^s && (78)\\
&= s\Pr[X \geq s] && (79)
\end{aligned}$$

and the reasoning is the same. □

Intuition. This simple result just states that, for a fixed, large t, the probability that a random variable is large is controlled by its expected value.

Theorem A.4 (Hoeffding's Lemma – symmetric interval, centered random variable)**.** *Let X be a centered random variable such that $|X| \leq a$ for some $a \in \mathbb{R}$. Then*

$$\mathbb{E}[e^{\lambda X}] \leq e^{\frac{\lambda^2 a^2}{2}} \tag{80}$$

$\forall \lambda \in \mathbb{R}$.

Proof. Since $X \mapsto e^{\lambda X}$ is a convex function, in the domain $[-a, a]$ one has

$$e^{\lambda X} \leq \frac{a+X}{2a}e^{\lambda a} + \frac{a-X}{2a}e^{-\lambda a}. \tag{81}$$

Taking expectations over X, we obtain

$$\begin{aligned}
\mathbb{E}[e^{\lambda X}] &\leq \mathbb{E}\left[\frac{a+X}{2a}e^{\lambda a} + \frac{a-X}{2a}e^{-\lambda a}\right]\\
&= \frac{a+\mathbb{E}[X]}{2a}e^{\lambda a} + \frac{a-\mathbb{E}[X]}{2a}e^{-\lambda a}\\
&= \frac{ae^{\lambda a} + ae^{-\lambda a}}{2a}\\
&= \frac{e^{\lambda a} + e^{-\lambda a}}{2}\\
&= \cosh(\lambda a). && (82)
\end{aligned}$$

Notice that the inequality is tight when X is a Rademacher random variable, that is when it has all the mass in the tails. Now, recall the Taylor series expansions

$$e^u = \sum_{k=0}^{\infty} \frac{u^k}{k!} \quad \Rightarrow \quad e^{u^2/2} = \sum_{k=0}^{\infty} \frac{(u^2/2)^k}{k!} = \sum_{k=0}^{\infty} \frac{u^{2k}}{2^k k!} \tag{83}$$

and

$$\cosh(u) = \sum_{k=0}^{\infty} \frac{u^{2k}}{(2k)!}. \tag{84}$$

It follows that to prove the result we just need to comapre $(2k)!$ and $2^k k!$. We have

$$\begin{aligned}
(2k)! &= (1 \cdot 3 \cdot 5 \cdots (2k-1)) \cdot (2 \cdot 4 \cdots 2k) \\
&= (1 \cdot 3 \cdot 5 \cdots (2k-1)) \cdot ((2 \cdot 1) \cdot (2 \cdot 2) \cdot (2 \cdot 3) \cdots (2 \cdot k)) \\
&= (1 \cdot 3 \cdot 5 \cdots (2k-1)) \cdot (2^k \cdot k!) \\
&\geq (2^k \cdot k!)
\end{aligned}$$

Hence, we obtain

$$\mathbb{E}[e^{\lambda X}] \leq \cosh(\lambda a) \leq e^{\lambda^2 a^2/2}. \tag{85}$$

□

Intuition. A naive intuition of Hoeffding's lemma is that any bounded random variable cannot fluctuate more than a Rademacher one. However, the result tells us more: it bounds the decay of the tail probability of a random variable with an exponential decay that is the Gaussian one. This is clear from the next version of the lemma and the following considerations.

Let us state, for the sake of completeness, a more general version of Hoeffding's lemma.

Theorem A.5 (Hoeffding's lemma)**.** *Let X be any random variable such that $a \leq X \leq b$ for some $a, b \in \mathbb{R}$. Then*

$$\mathbb{E}[e^{\lambda X}] \leq e^{\lambda \mathbb{E}[X] + \frac{\lambda^2 (b-a)^2}{8}} \tag{86}$$

$\forall \lambda \in \mathbb{R}$.

Proposition A.6 (Gaussian tail bound)**.** *The Gaussian tail bound used above to obtain (6) comes from the following result. Consider a generic Gaussian integral over the rightmost tail, then*

$$\int_{x_0}^{\infty} e^{-cx^2} \, \mathrm{d}x \leq \frac{1}{2x_0 c} e^{-cx_0} \tag{87}$$

Proof. We express the integrand as the derivative of an exponential, and then use IBP

$$\begin{aligned}
\int_{x_0}^{\infty} e^{-cx^2} \, \mathrm{d}x &= \int_{x_0}^{\infty} -\frac{1}{2cx} \frac{\mathrm{d}e^{-cx^2}}{\mathrm{d}x} \, \mathrm{d}x = -\frac{e^{-cx^2}}{2cx}\bigg|_{x_0}^{\infty} - \int_{x_0}^{\infty} \frac{\mathrm{d}}{\mathrm{d}x}(-\frac{1}{2cx}) e^{-cx^2} \, \mathrm{d}x \\
&= \frac{1}{2cx_0} e^{-2cx_0} - \int_{x_0}^{\infty} \frac{\mathrm{d}}{\mathrm{d}x}(-\frac{1}{2cx}) e^{-cx^2} \, \mathrm{d}x \leq \frac{1}{2cx_0} e^{-2cx_0}
\end{aligned}$$

where the inequality follows from

$$\frac{\mathrm{d}}{\mathrm{d}x}(-\frac{1}{cx^2}) \geq 0 \tag{88}$$

and hence the integral being non-negative. □

Remark A.7 (Hoeffding's lemma and subgaussian random variables)**.** *A centered random variable is said subgaussian with parameter γ if*

$$\mathbb{E}[e^{\lambda X}] \leq e^{\frac{\lambda^2\gamma^2}{2}}, \tag{89}$$

for any $\lambda \in \mathbb{R}$. The definition comes from the fact that for a centered, gaussian random variable, the inequality is sharp with $\gamma = \sigma$ standard deviation of X. Hoeffding's lemma just says that any centered random variable bounded as $|X| \leq a$ is subgaussian with parameter a. Also, its general version says that if $a \leq X \leq b$, then X is subgaussian with parameter $\frac{b-a}{2}$.

The next two results are the most *operational* ones, which aim at bounding the deviation of a sum of random variables from their mean. They follow directly from Theorems A.3 and A.4.

Theorem A.8 (Hoeffding's inequality)**.** *Let $Z_1, \dots, Z_n$ be independent real-valued random variables such that $\forall i$, $Z_i \in [a_i, b_i]$ almost surely for some $a_i, b_i \in \mathbb{R}$, and let $\mu := \mathbb{E}\left[\frac{1}{n}\sum_i Z_i\right] = \frac{1}{n}\sum_i \mathbb{E}[Z_i]$. Then, for every $\varepsilon > 0$,*

$$\Pr\left[\frac{1}{n}\sum_{i=1}^{n} Z_i - \mu \geq \varepsilon\right] \leq \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right). \tag{90}$$

non c'è un abs value? come so qual è maggiore tra empirical mean e μ?

Proof. Let $Z_n := \sum_i Z_i$. We will prove the equivalent statement

$$\Pr\left[Z_n - \mathbb{E}[Z_n] \geq \varepsilon\right] \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right). \tag{91}$$

By Markov's inequality applied to $e^{t(\sum_i Z_i - \mathbb{E}[\sum_i Z_i])}$ and independence of the Z_i, $\forall \lambda > 0$ it holds

$$\Pr\left[Z_n - \mathbb{E}[Z_n] \geq \varepsilon\right] = \Pr\left[e^{\lambda\sum_i (Z_i - \mathbb{E}[Z_i])} \geq e^{\lambda\varepsilon}\right] \leq \frac{\mathbb{E}\left[e^{\lambda\sum_i (Z_i - \mathbb{E}[Z_i])}\right]}{e^{\lambda\varepsilon}} = e^{-\lambda\varepsilon}\prod_{i=1}^{n}\mathbb{E}\left[e^{\lambda(Z_i - \mathbb{E}[Z_i])}\right]. \tag{92}$$

By Hoeffding's lemma, since $Z_i - \mathbb{E}[Z_i]$ has zero mean and takes values in an interval of length $b_i - a_i$, we have

$$\mathbb{E}\left[e^{\lambda(Z_i - \mathbb{E}[Z_i])}\right] \leq \exp\left(\frac{\lambda^2(b_i - a_i)^2}{8}\right), \tag{93}$$

so that

$$\Pr\left[Z_n - \mathbb{E}[Z_n] \geq \varepsilon\right] \leq \exp\left(-\lambda\varepsilon + \frac{\lambda^2}{8}\sum_i (b_i - a_i)^2\right). \tag{94}$$

The right-hand side is minimized at $\lambda = 4\varepsilon/\sum_i (b_i - a_i)^2$. Substituting this value yields

$$\Pr\left[Z_n - \mathbb{E}[Z_n] \geq \varepsilon\right] \leq \exp\left(-\frac{2\varepsilon^2}{\sum_i (b_i - a_i)^2}\right). \tag{95}$$

□

Theorem A.9 (Bernstein's inequality)**.** *Let $X_1, \dots, X_n$ be independent real-valued random variables such that $\forall i$, $\mathbb{E}[X_i] = 0$ and $X_i \leq M$ almost surely for some $M \in \mathbb{R}$. Let $\sigma^2 = \frac{1}{n}\sum_i \mathrm{Var}(X_i)$. Then, for every $\varepsilon > 0$,*

$$\Pr\left[\frac{1}{n}\sum_{i=1}^{n} X_i > \varepsilon\right] \leq e^{-\frac{n\varepsilon^2}{2(\sigma^2 + M\varepsilon/3)}}. \tag{96}$$

Proof. Consider the moment generating functino (MGF) $e^{\lambda X_i}$. We have

$$e^{\lambda X_i} = 1 + \lambda e^{\lambda X_i} + \frac{1}{2}\lambda^2 X_i^2 + \frac{1}{3!}\lambda^3 X_i^3 + \frac{1}{4!}\lambda^4 X_i^4 + \cdots . \tag{97}$$

Taking expectations

$$\mathbb{E}[e^{\lambda X_i}] = 1 + \frac{1}{2}\lambda^2 \mathbb{E}[X_i^2] + \frac{1}{3!}\lambda^3 \mathbb{E}[X_i^3] + \frac{1}{4!}\lambda^4 \mathbb{E}[X_i^4] + \cdots . \tag{98}$$

Then, since $X_i \leq M$,

$$\frac{X_i}{M} \leq 1 \qquad \Rightarrow \qquad \left(\frac{X_i}{M}\right)^r \leq \left(\frac{X_i}{M}\right)^2 \quad \forall r \geq 2 \qquad \Rightarrow \qquad X_i^r \leq X_i^2 M^{r-2}. \tag{99}$$

Hence from equation (98),

$$\mathbb{E}[e^{\lambda X_i}] = 1 + \frac{1}{2}\lambda^2 \mathbb{E}[X_i^2] + \frac{1}{3!}\lambda^3 \mathbb{E}[X_i^3] + \frac{1}{4!}\lambda^4 \mathbb{E}[X_i^4] + \cdots \tag{100}$$

$$\leq 1 + \frac{\mathbb{E}[X_i^2]}{M^2}\left(\frac{1}{2}\lambda^2 M^2 + \frac{1}{3!}\lambda^3 M^3 + \frac{1}{4!}\lambda^4 M^4 + \cdots\right) \tag{101}$$

$$= 1 + \frac{\mathbb{E}[X_i^2]}{M^2}(e^{\lambda M} - 1 - \lambda M) \tag{102}$$

$$= 1 + \frac{\mathrm{Var}[X_i]}{M^2}(e^{\lambda M} - 1 - \lambda M) \tag{103}$$

Let us define $e^x - 1 - x =: \psi(x)$. Using that $1 + x \leq e^x$, one obtains

$$\mathbb{E}[e^{\lambda X_i}] \leq 1 + \frac{\psi(\lambda M)}{M^2}\mathrm{Var}[X_i] \leq \exp\left(\frac{\psi(\lambda M)}{M^2}\mathrm{Var}[X_i]\right). \tag{104}$$

- chernoff
- optimizing over λ
- elementary inequality □

Intuition. The main ingredients of these proof are two:

- first, we Taylor expand the MGF and set to 0 the first moment, since our random variable is centered;
- then, since our random variable is bounded, we upper bound higher-order moments with the second-order one, since if $0 \leq p \leq 1, p^r \leq p^2 \forall r \geq 2$, and we add a correction term for our upper bound M possibly greater than 1.

These two ingredients, together, allow to bound the MGF of X with its variance, adding more of the structure of the problem to our bounds.

Lemma A.10 (Young's inequality)**.** *Let $a, b \geq 0$ and let $p, q > 1$ be conjugate exponents, i.e.*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. Fix $b \geq 0$ and consider

$$\phi(a) = \frac{a^p}{p} - ab + \frac{b^q}{q}, \qquad a \geq 0.$$

Since

$$\phi'(a) = a^{p-1} - b \qquad \text{and} \qquad \phi''(a) = (p-1)a^{p-2} \geq 0,$$

ϕ attains its minimum at

$$a = b^{1/(p-1)} = b^{q-1}.$$

Using $q = p/(p-1)$, at this point

$$\phi(a) = \frac{b^q}{p} - b^q + \frac{b^q}{q} = b^q \left(\frac{1}{p} + \frac{1}{q} - 1 \right) = 0.$$

Therefore $\phi(a) \geq 0$ for all $a \geq 0$, which is equivalent to

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

□

Appendix B. Toy example

Example B.1 (Least squares with toy dataset)**.** *Let $\ell(f(x_i), y_i) = (f(x_i) - y_i)^2$ the quadratic loss, and $\mathcal{H} = \{x \mapsto ax : a \in \mathbb{R}\}$ the space of linear functions. We obtain the least squares estimator*

$$\hat{f}_d(x_i) = \frac{\sum_i x_i y_i}{\sum_i x_i^2} x_i, \qquad x_i, y_i \in d. \tag{105}$$

Now, consider $d = \{(x_1, y_1), (x_2, y_2)\}$ realization of D. First, we fit our estimator to the observation (x_1, y_1) and use (x_2, y_2) as a test set, in the spirit of classical statistical theory. The empirical risk equals

$$\begin{aligned} \hat{R}_{d_2}(\hat{f}_{d_1}) &= \ell(\hat{f}_{d_1}(x_2) - y_2) \\ &= \left(\frac{x_1 y_1}{x_1^2} x_2 - y_2\right)^2 \\ &= \frac{(x_1 y_1)^2}{x_1^4} x_2^2 + y_2^2 - 2 x_2 y_2 \frac{x_1 y_1}{x_1^2} \end{aligned} \tag{106}$$

Taking expectations (with a little abuse of notation),

$$\begin{aligned} \mathbb{E}_{d_2 \sim p}[\hat{R}_{d_2}(\hat{f}_{d_1})] &= \sum_{x_2, y_2} \ell(\hat{f}_{d_1}(x_2) - y_2) p(x_2, y_2) \\ &= \frac{(x_1 y_1)^2}{x_1^4} \sum_{x_2} x_2^2 p_x(x_2) + \sum_{y_2} y_2^2 p_y(y_2) - 2 \frac{x_1 y_1}{x_1^2} \sum_{x_2, y_2} p(x_2, y_2) x_2 y_2 \end{aligned} \tag{107}$$

where $\sum_x$ is the sum over all possible values of x, and p_x, p_y are marginal distributions. Recall that the expected risk equals

$$R(\hat{f}_{d_1}) = \sum_{s_i, y_i} \ell(\hat{f}_{d_1}(s_i) - y_i) p(s_i, y_i) \tag{108}$$

and since the sums in equation (107) run over all the possible values x_2 and y_2 can take, the two quantities are equal. The result does not change if we consider larger datasets. Now, we look at the case where we fit and evaluate the estimator on a single dataset (x_1, y_1). In this case, we would have

$$\mathbb{E}[\hat{R}_{d_1}(\hat{f}_{d_1})] = \sum_{x_1, y_1} \frac{(x_1 y_1)^2}{x_1^4} x_1^2 p_x(x_1) + \sum_{y_1} p_y(y_1) y_1^2 - \sum_{x_1, y_1} 2 \frac{x_1 y_1}{x_1^2} p(x_1, y_1) x_1 y_1. \tag{109}$$

Where the "cross" terms make the estimator biased. Doing the math,

$$\mathbb{E}[\hat{R}_{d_1}(\hat{f}_{d_1})] = 2 \sum_{y_1} y_1^2 p_y(y_1) - 2 \sum_{y_1} y_1^2 p_y(y_1) = 0 \leq R(\hat{f}_{d_1}) \tag{110}$$

where the inequality comes from $\ell(\cdot) \geq 0$. Hence, in this example the empirical risk evaluated on the same data to which the estimator is fitted underestimates the risk of the estimator. We will see that this is always the case.

Appendix C. Bound proposition in different settings

Here, we show the bound result Theorem 2.5 in two different seetings from before.

Sign function and quadratic loss. Let $\mathcal{G}$ be the space of functions $g : \mathcal{X} \to \mathbb{R}$ and $\mathcal{H}$ the space of functions $f : \mathbb{R} \to \mathcal{Y}$ such that

$$f \coloneqq \operatorname{sign} \circ g. \tag{111}$$

This means that for any g there is a $f_g = \operatorname{sign} \circ g$. Observe that g encodes the complexity of our hypotheses space (e.g. the structure of the tree in the case of decision trees) and f encodes binary predictions. Notice that if the size of the support is $|\mathcal{X}| = N$, then $|\mathcal{H}| = 2^N$. Also, we consider

$$\ell(f(x_i), y_i) = (y_i - f(x_i))^2 \tag{112}$$

the quadratic loss.

Proposition C.1 (Bounds on the bias I)**.** *Given data $D : \Omega^n \to \mathcal{X}^n \times \mathcal{Y}^n$ and a predictor $f : \mathcal{X} \to \{-1, 1\}$, the empirical and the expected risk of trained and evaluated on D with a quadratic loss function satisfy the following bound*

$$0 \leq \mathbb{E}_D[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D)] \leq 4\sqrt{\frac{2 \log |\mathcal{H}|}{n}}. \tag{113}$$

Proof. We prove the lower bound ($0 \leq \mathbb{E}_D \ldots$) first. All the expectations are over the randomness of the data D, we drop the subscript to light notation. We have

$$\hat{R}_D(\hat{f}_D) \leq \hat{R}_D(f) \quad \forall f \in \mathcal{H}, \tag{114}$$

since $\hat{f}_D \coloneqq \arg\min_{f \in \mathcal{H}} \hat{R}_D(f)$. Then, take expectations

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq \mathbb{E}[\hat{R}_D(f)] \quad \forall f \in \mathcal{H}. \tag{115}$$

Now, since the functions in $\mathcal{H}$ are deterministic, by Proposition 1.10 we have

$$\mathbb{E}[\hat{R}_D(f)] = R(f) \quad \forall f \in \mathcal{H}, \tag{116}$$

so that

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq R(f) \quad \forall f \in \mathcal{H}. \tag{117}$$

Now take d a generic realization of D and consider in particular $f = \hat{f}_d$,

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq R(\hat{f}_d) \quad \forall d \in \mathcal{D}. \tag{118}$$

Since this is valid pointwise for any d realization of D, we have

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq R(\hat{f}_D) \quad \text{a.s.} \tag{119}$$

and, finally, taking expectations

$$\mathbb{E}[\hat{R}_D(\hat{f}_D)] \leq \mathbb{E}[R(\hat{f}_D)] \quad \text{a.s.} \tag{120}$$

Next, we prove the upper bound ($\mathbb{E} \leq 2\sqrt{\ldots}$). For any $f \in \mathcal{H}$, $\mathbb{E}[\hat{R}_D(f)] = R(f)$. Let $Z_i \coloneqq \ell(f(x_i), y_i)$ and have

$$\hat{R}_D(f) = \frac{1}{n} \sum_i Z_i \tag{121}$$

We now want to apply Hoeffding's inequality. To do so, we need to satisfy 3 conditions:

(*i*) $Z_1, \dots, Z_n$ i.i.d.

(*ii*) $\mathbb{E}[Z_1] = \cdots = \mathbb{E}[Z_n]$

(*iii*) $b_i \leq Z_i \leq a_i$ for some $a_1, \dots, a_n \in \mathbb{R}$ and $b_1, \dots, b_n \in \mathbb{R}$.

We have that (*i*) is satisfied by $Z_i s$ being i.i.d. by construction. Condition (*ii*) is satisfied since all the (x_i, y_i) have the same distribution. As for (*iii*), it is satisfied by the structure of the problem (quadratic loss) with $b_1 = \cdots = b_n = 0$ and $a_1 = \cdots = a_n = 4$. Now we can apply Hoeffding's. We obtain

$$\Pr[\frac{1}{n}\sum_i Z_i - \mu \geq \varepsilon] \leq e^{-n\varepsilon^2/8}$$

and thus

$$\Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] \leq e^{-n\varepsilon^2/8} \tag{122}$$

Now, to obtain a bound for $\hat{f}_D$ we bound it with the union of the probabilities of the events for each $f \in \mathcal{H}$. We have

$$\Pr[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D) \geq \varepsilon] \leq \bigcup_{f \in \mathcal{H}} \Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] \tag{123}$$

trivially, and also

$$\bigcup_{f \in \mathcal{H}} \Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] \leq \sum_{f \in \mathcal{H}} \Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] \leq \sum_{f \in \mathcal{H}} e^{-n\varepsilon^2/8} = |\mathcal{H}| e^{-n\varepsilon^2/8} \tag{124}$$

where the first inequality follows by a union bound and the second by Hoeffding. All in all, we obtain

$$\Pr[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D) \geq \varepsilon] \leq |\mathcal{H}| e^{-n\varepsilon^2/8} \tag{125}$$

To obtain our initial inequality from this result on a probability, we use a technique called tail integration. The key ingredient is the following

$$\mathbb{E}[X] = \int_0^\infty \Pr(X \geq \varepsilon) \ \mathrm{d}\varepsilon \tag{126}$$

which is true for any non-negative random variable X, see Proposition A.2 in the appendix. Then, we want to use our bound (125) for the integrand. Let $a \coloneqq |\mathcal{H}|$ and $c \coloneqq n/8$, so that

$$\Pr(X \geq \varepsilon) \leq a e^{-c\varepsilon^2}. \tag{127}$$

Notice that for high values of ε, the bound is a very good one, but for $\varepsilon \gg 1$ it is bad. Indeed, we can use a better bound:

$$\Pr[X \geq \varepsilon] \leq \min\{1, a e^{-c\varepsilon^2}\} \tag{128}$$

Therefore, it makes sense to split the integral in (126) in two parts with a threshold $a e^{-c\varepsilon_0^2} = 1$. This means splitting the integral at $\varepsilon_0 = \sqrt{\frac{\log a}{c}}$. We obtain

$$\begin{aligned}\mathbb{E}[X] &= \int_0^\infty \Pr(X \geq \varepsilon) \ \mathrm{d}\varepsilon \leq \int_0^{\sqrt{\frac{\log a}{c}}} 1 \ \mathrm{d}\varepsilon + \int_{\sqrt{\frac{\log a}{c}}}^\infty a e^{-c\varepsilon^2} \ \mathrm{d}\varepsilon \\ &= \sqrt{\frac{\log a}{c}} + a \frac{1}{2\sqrt{c \log a}} e^{-c\varepsilon^2} = \sqrt{\frac{\log a}{c}} + \frac{1}{2\sqrt{c \log a}}\end{aligned}$$

where the second integral is solved with a standard Gaussian tail bound (see Proposition A.6), and in the last equality we substituted $a = 1/e^{-c\varepsilon^2}$. Finally, we can use a looser but simpler bound assuming $a \geq e^{1/2}$, which is sensible since $a = |\mathcal{H}|$ and $e^{1/2} < 2$,

$$\mathbb{E}[X] \leq 2\sqrt{\frac{\log a}{c}} \tag{129}$$

which yields the result for $X = R(\hat{f}_D) - \hat{R}(\hat{f}_D)$, $a = |\mathcal{H}|$, and $c = n/8$. □

Remark C.2 (Logic of the proof)**.** *Intuitively, the key step in the proof of the lower bound is that inequality (117) is an inequality between numbers. The expected value of $\hat{R}_D(\hat{f}_D)$ is a number, say 3, and we are showing that this is lower than all the numbers $R(f), f \in \mathcal{H}$. Instead, in inequality (122) we use a probabilistic bound and work with random variables.*

Real-valued prediction function and logistic loss. Now, we study the same result but in a different setting, namely considering real-valued prediction functions instead of a sign function, and a logistic loss. Let $\mathcal{H}$ be the space of functions $f : \mathcal{X} \to \mathbb{R}$, for instance linear functions of the type

$$f_{wb}(x) \coloneqq wx + b, \qquad w, b \in \mathbb{R}. \tag{130}$$

Notice that if $\mathcal{X}$ has size N, then $\mathcal{H} = \{\text{sign}(f(x)) | w, b \in \mathbb{R}\}$ is such that $|\mathcal{H}| = 2(N+1)$. Furthermore, we consider the logistic loss

$$\ell(f(x_i), y_i) = \log(1 + e^{-y_i f(x_i)}). \tag{131}$$

Then, the following result holds on the bias between the expected and empirical risks.

Proposition C.3 (Bounds on the bias II)**.** *Given data $D : \Omega^n \to \mathcal{X}^n \times \mathcal{Y}^n$ and a predictor $f : \mathcal{X} \to \mathbb{R}$, the empirical and the expected risk of trained and evaluated on D with a logistic loss function satisfy the following bound*

$$0 \leq \mathbb{E}_D[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D)] \leq 2\sqrt{\frac{\log |\mathcal{H}|}{n}} \tag{132}$$

Proof. The proof of the lower bound is the same as in Proposition E.2. We therefore show the proof of the upper bound ($\mathbb{E} \leq \ldots$). Again, we define $Z_i \coloneqq \ell(f(x_i), y_i)$ and check the three conditions required by Hoeffding's inequality. As before, the random variables are i.i.d. and have the same expected value by construction. As for obtaining bounds on Z_i, we perform the following rescaling for $f_{w,b}(x) \neq 0$

$$f_{w,b}(x) = \frac{f_{w,b}(x)}{\|f_{w,b}\|_\infty} \tag{133}$$

where

$$\|f\|_\infty \coloneqq \max_{x \in \mathcal{X}} |f_{w,b}(x)|. \tag{134}$$

Then, we obtain

$$-1 \leq f(x_i) \leq 1 \qquad \forall i \tag{135}$$

and hence

$$\log(1 + \frac{1}{e}) \leq \log(1 + e^{-y_i f(x_i)}) \leq \log(1 + e) \qquad \forall i. \tag{136}$$

Plugging these bounds into Hoeffding's inequality, we obtain

$$\Pr[\frac{1}{n}\sum_i Z_i - \mu \geq \varepsilon] \leq e^{-n\varepsilon^2/8} \tag{137}$$

and thus

$$\begin{aligned}
\Pr[R(f) - \hat{R}_D(f) \geq \varepsilon] &\leq e^{-n\varepsilon^2/(\log(1+e) - \log(1+\frac{1}{e}))^2} \\
&= e^{-n\varepsilon^2/(\log(\frac{1+e}{1+\frac{1}{e}}))^2} \\
&= e^{-n\varepsilon^2/(\log e)^2} \\
&= e^{-n\varepsilon^2}
\end{aligned}$$

Now, proceeding as before (trivial bound with the union + union bound) we obtain

$$\Pr[R(\hat{f}_D) - \hat{R}_D(\hat{f}_D) \geq \varepsilon] \leq |\mathcal{H}| e^{-n\varepsilon^2}. \tag{138}$$

The proof is then completed exactly as before, with parameters $a = |\mathcal{H}|$ and $c = n$. □

Appendix D. Additional Proofs

Remark D.1 (Additional notes on variance-based bound)**.** *Recall inequality (45) from Section 2.2. We first discuss why it is an improvement. If* $\mathrm{Var}(f) \geq \varepsilon$*, the variance controls the behavior of the denominator* $2(\mathrm{Var}(f)+M\varepsilon/3)$*; instead, if* $\mathrm{Var}(f) \leq \varepsilon$*, i.e. if the variance is small, the denominator is* $2(\mathrm{Var}(f)+M\varepsilon/3) \leq \frac{2}{3}(3+M)\varepsilon$ *and the upper bound behaves like* $e^{-n\varepsilon}$ *instead of* $e^{-n\varepsilon^2}$ *as in inequality (44)* [3]*. Moreover, notice that*

$$\begin{aligned}
\mathrm{Var}(f) &= \mathrm{Var}(\ell(f(X), Y)) && (139)\\
&= \frac{1}{n}\sum_i \mathrm{Var}(\ell(f(X_i), Y_i)) && (140)\\
&= n \cdot \frac{1}{n^2}\sum_i \mathrm{Var}(\ell(f(X_i), Y_i)) && (141)\\
&= n\,\mathrm{Var}\left(\frac{1}{n}\sum_i \ell(f(X_i), Y_i)\right) && (142)\\
&= n\,\mathrm{Var}(\hat{R}_D(f)). && (143)
\end{aligned}$$

When ℓ *is the counting loss,* $\hat{R}_D(f)$ *is a Binomial random variable whose variance is* $R(f)(1-R(f)) \leq R(f)$*. Hence,* $\mathrm{Var}(f)$ *is small when the expected risk is small, and in this case (45) is much better than (44).*

Next, we show that our inequality (45) is equivalent to inequality (13) *in* [2]*, Section* 5*. As a consequence of (45),*

$$\Pr[R(f) - \hat{R}_D(f) \leq \varepsilon] \geq 1 - e^{-\frac{n\varepsilon^2}{2(\mathrm{Var}(f)+M\varepsilon/3)}}. \tag{144}$$

Take $\delta \in \mathbb{R}^+$*. We want to find for which value of* ε*,* $1-\delta$ *is a lower bound on the probability that the event is true, hence* $1 - \delta \leq 1 - e^{-\frac{n\varepsilon^2}{2(\mathrm{Var}(f)+M\varepsilon/3)}}$*. We have*

$$\begin{aligned}
\delta &\geq e^{-\frac{n\varepsilon^2}{2(\mathrm{Var}(f)+M\varepsilon/3)}}; && (145)\\
\log\delta &\geq -\frac{n\varepsilon^2}{2(\mathrm{Var}(f)+M\varepsilon/3)}; && (146)\\
2\log\frac{1}{\delta}(\mathrm{Var}(f)+M\varepsilon/3) &\leq n\varepsilon^2; && (147)\\
n\varepsilon^2 - \frac{2}{3}M\log\frac{1}{\delta}\varepsilon &\geq 2\log\frac{1}{\delta}\mathrm{Var}(f) && (148)\\
\varepsilon^2 - \frac{2}{3n}M\log\frac{1}{\delta}\varepsilon &\geq \frac{2\log\frac{1}{\delta}\mathrm{Var}(f)}{n} && (149)
\end{aligned}$$

. . . finisci

Appendix E. Mean analysis

Qui, qualche appunto con l'obiettivo di indagare

quanto peggiora la stima dell'errore di un modello sul train set al crescere del numero di modelli considerati?

Notation and references. We start from the notes 'ML ideas' by Prof. Rosasco. We consider the following items:

- a probability distribution p over $S \subseteq \mathbb{R}$;
- support of the distribution $s = s_1, \dots, s_N$, indexed with j, with probabilities p_j:

$$\mathbb{E}(X) = \sum_j s_j p_j =: \mu; \qquad \mathrm{Var}(X) = \sum_j s_j^2 p_j - (\sum_j s_j p_j)^2 =: \sigma^2; \tag{150}$$

- random variables $X_1, \dots, X_n$, i.i.d.'to $X \sim p(\mu, \sigma^2)$;
- a dataset $x_1, \dots, x_n$ realization of $X_1, \dots, X_n$, indexed with i, which is therefore assumed to be sampled from our probability distribution p (the assumption that Ben Recht doesn't like).

We aim at obtaining predictions from our data. To be able to evaluate predictors, we define the following measure of error of a scalar predictor a.

Definition E.1 (Expected risk)**.** *The expected risk of a predictor $a \in \mathbb{R}$ is:*

$$R(a) \coloneqq \sum_j (s_j - a)^2 p_j \tag{151}$$

The motivation for this definition is that we aim at predicting some value a from our distribution, and the best prediction is the one that is closest (in terms of squared distance) to each possible realization of the distirbution, weighted by their masses.

Proposition E.2. *The following results hold*

(i) by realizing the distirbution p as our random varibale X, the expected risk can be expressed as

$$R(a) = \mathbb{E}[(X - a)^2] \tag{152}$$

(ii)

$$R(a) = (a - \mu)^2 + \sigma^2 \tag{153}$$

Proof. (i) follows plugging in $\sum s_j p_j = \mathbb{E}[X]$ and from a being a scalar:

$$\begin{aligned} R(a) &= \sum_j (s_j - a)^2 p_j = \sum_j (s_j^2 + a^2 - 2 s_j a) p_j = \sum_j s_j^2 p_j + \sum_j a^2 p_j - 2 \sum_j s_j a p_j \\ &= \mathbb{E}[X^2] + a^2 - 2a\mathbb{E}[X] = \mathbb{E}[X^2 - 2aX + a^2] = \mathbb{E}[(X - a)^2] \end{aligned}$$

and to obtain (ii) we add and subtract $\mathbb{E}^2[X]$:

$$R(a) = (\mathbb{E}[X^2] - \mathbb{E}^2[X] + \mathbb{E}[X^2] + a^2 - 2a\mathbb{E}[X]) = \sigma^2 + \mu^2 - 2a\mu + a^2 = \sigma^2 + (\mu - a)^2$$

□

In practice, the expected risk cannot be computed as we do not know the true distribution p. Therefore we use the following empirical error measure.

Definition E.3 (Empirical risk)**.** *The empirical risk of a predictor $a \in \mathbb{R}$ for some data $x_1, \dots, x_n$ is*

$$\hat{R}(a) \coloneqq \frac{1}{n} \sum_i (x_i - a)^2 \tag{154}$$

A first result on the empirical risk is that it is an *unbiased estimate of the expected risk*. Let us pause for a moment here. When evaluating the unbiasedness of the empirical risk, implicitamente stiamo dicendo che l'errore che un estimatore fa sui dati è aleatorio. Questa aleatorietà viene dalla aleatorietà dei dati (l'unica fonte, essendo per ora a uno scalare). Infatti, il rischio empirico si può riformulare come

$$\hat{R}(a) = \frac{1}{n}\sum_i (X_i - a)^2 \qquad \text{v.s.} \qquad R(a) = \mathbb{E}[(X-a)^2] \tag{155}$$

Quindi, noi abbiamo una distribuzione di probabilità (popolazione) su cui l'errore di un estimatore è costante (e si può calcolare se è nota p, entropia 0); poi, abbiamo dei dati che sono campionati da questa distribuzione (ben recht assumption), che sono modellati come una collezione di variabili aleatorie, su cui l'errore dunque Ma se l'errore sulla distribuzione è costante, come può non esserlo l'errore sui dati campionati da quella distribuzione? Qui che c'è un key point concettuale: potenzialmente, se io campionassi solo due dati x_1, x_2, uno scalare che ha expected risk altissimo potrebbe essere top in termini di empirical risk perché per *pura casualità* è un numero vicino ai miei dati x_1, x_2. Quindi il rischio atteso mi definisce il rischio che *mi aspetto* il mio estimatore abbia data quella distribuzione. Di fatto, non solo non lo posso calcoalre, ma neanche mi interessa troppo calcolarlo, perché se avessi solo due osservazioni non mi interesserebbe nulla del rischio atteso, guarderei solo al rischio empirico. Se invece ho tante osservazioni, allora il rischio atteso e quello empirico si allineano, perché i miei dati andranno a copiare la distribuzione (concetto di expectation / long run). Questo si formalizza nella seguente prop.

Proposition E.4 (Unbiasedeness of the empirical risk)**.** *For a scalar predictor $a \in \mathbb{R}$, the empirical risk is an unbiased estimate of the expected risk, i.e.*

$$\mathbb{E}[\hat{R}(a)] = R(a). \tag{156}$$

Proof.

$$\mathbb{E}[\hat{R}(a)] = \mathbb{E}[\frac{1}{n}\sum_i (X_i - a)^2] = \frac{1}{n}\sum_i \mathbb{E}[(X_i - a)^2] = \frac{1}{n} n \cdot \mathbb{E}[(X_i - a)^2] = \mathbb{E}[(X_i - a)^2] = R(a) \tag{157}$$

using Proposition E.2 (i) and that $X_1, \dots, X_n$ are i.i.d. this is the same motivation behind the empriical mean being an unbiased estimator of the true mean. □

di base, la nozione di empirical risk viene dalla seguente intuizione: siccome non posso conoscere le varie probabilità p_j, le stimo come il numero di volte che la mia variabile aleatoria (campione) prendere quel valore $s_j \leftarrow$ legge dei grandi numeri

Remark E.5. *usiamo in tutte queste note (come in letteratura) la stessa notazione per una quantità aleatoria espressa in termini di X_i e la sua realizzazione espressa in termini di x_i (come per $\hat{R}(a)$ above). Questo perché X e x possono essere essenzialmente lo stesso oggetto, cioè X è una funzione la cui immagine è x*

una volta consolidata questa teoria e il rapporto tra rischio atteso e rischio empirico, guardiamo al caso in cui a non è uno scalare.

per esempio, assumiamo di voler stimare la media della nostra distribuzione. Vogliamo farlo anche perché è chiaro che il valore costante che minimizza il rischio atteso è la media della distribuzione, e il valore costante che minimizza il rischio empirico è la media empirica dei dati (essenzialmente by definition, la media è il valore che minimizza la distanza!).

soffermiamoci per un momento anche sull'idea che la media empirica minimizza il rischio empirico. Come si relaziona questo al voler prevedere un nuovo dato x_{n+1}? la teoria mi starebbe dicendo che se io prevedo la media dei dati che ho, alla fine avrò la perdita minore possibile. In realtà però a me converrebbe prevedere

il dato che vedo apparire più spesso, naivly, se il mio obiettivo è azzeccare il next sample. Ma forse il mio obiettivo è avere un costo basso overall.

ora, la teoria mi dice di prevedere la media, ma io voglio quantificare quanto è un buon stimatore questa media. Perché so che minimizza il rischio empirico, ma what about il rischio atteso? ecco, nel caso della sample mean, il rischio empirico è un biased estimate del rischio atteso. Vediamolo:

Proposition E.6 (Biasedeness of the empirical risk for $\hat{\mu}$)**.** *The empirical risk $\hat{R}$ is a biased estimate of the true expected risk R for the empirical mean $\hat{\mu} = \frac{1}{n}\sum_i x_i$.*

Proof. The object we are looking at is

$$\hat{R}(\hat{\mu}) \coloneqq \frac{1}{n}\sum_{i=1}^{n}(X_i - \hat{\mu})^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \frac{1}{n}\sum_{j=1}^{n}X_j)^2 \tag{158}$$

and

$$R(\hat{\mu}) \coloneqq \mathbb{E}[(X - \hat{\mu})^2] = \mathbb{E}[(X - \frac{1}{n}\sum_{i=1}^{n}X_i)^2] \tag{159}$$

and we recall that we should look at $\hat{R}$ as a "score" we give to the estimator for the mean, the lower the better, while R is the true score of the estiimator. We have

$$\begin{aligned}
\mathbb{E}[\hat{R}(\hat{\mu})] &= \mathbb{E}[\frac{1}{n}\sum_{i=1}^{n}(X_i - \frac{1}{n}\sum_{j=1}^{n}X_j)^2] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i - \frac{1}{n}\sum_{j=1}^{n}X_j)^2 \\
&= \frac{1}{n}\sum_{i=1}^{n}(\mathbb{E}X_i^2 + \frac{1}{n^2}\mathbb{E}(\sum_{j=1}^{n}X_j)^2 - \frac{2}{n}\mathbb{E}(X_i\sum_{j=1}^{n}X_j)) \\
&= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}X_i^2 + \frac{1}{n^2}\mathbb{E}(\sum_{j=1}^{n}X_j)^2 - \frac{1}{n}\sum_{i=1}^{n}\frac{2}{n}\mathbb{E}(X_i\sum_{j=1}^{n}X_j)
\end{aligned}$$

and, similarly,

$$R(\hat{\mu}) = \mathbb{E}X^2 + \frac{1}{n^2}\mathbb{E}(\sum_{i=1}^{n}X_i)^2 - \frac{2}{n}\mathbb{E}(X\sum_{i=1}^{n}X_i) \tag{160}$$

The orange terms are equal by $X_i \sim X\ \forall i = 1, \dots, n$, the blue terms are identical, hence we study the pink terms to evaluate the bias.

$$-\frac{2}{n}\mathbb{E}(X\sum_{i=1}^{n}X_i) = -\frac{2}{n}\sum_i\mathbb{E}(XX_i) = -\frac{2}{n}\sum_i\mathbb{E}(X)\mathbb{E}(X_i) = -\frac{2}{n}n\mathbb{E}^2(X) = -2\mathbb{E}^2(X) \tag{161}$$

because X, X_i are independent, and $\mathbb{E}X_i = \mathbb{E}X$. As for the empirical risk, we have

$$\begin{aligned}
&-\frac{1}{n}\sum_i\frac{2}{n}\mathbb{E}(X_i\sum_j X_j) \\
&= -\frac{2}{n^2}[\mathbb{E}(X_1^2 + X_1X_2 + \cdots + X_1X_n) + \mathbb{E}(X_2X_1 + X_2^2 + \cdots + X_2X_n) + \cdots + \mathbb{E}(X_nX_1 + \cdots + X_n^2)] \\
&= -\frac{2}{n^2}[\sum_i\mathbb{E}(X_i^2) + \sum_{i\neq j}\mathbb{E}(X_i)\mathbb{E}(X_j)] = -\frac{2}{n^2}n\mathbb{E}(X^2) - \frac{2}{n^2}(n^2-n)\mathbb{E}^2(X) = -\frac{2}{n}\mathbb{E}(X^2) - (2 - \frac{2}{n})\mathbb{E}^2(X)
\end{aligned}$$

where the coefficients $-\frac{2}{n} - 2 + \frac{2}{n}$ sum to -2, but crucially

$$\mathbb{E}(X^2) \neq \mathbb{E}^2(X) \tag{162}$$

because $\mathbb{E}(X^2) - \mathbb{E}^2(X) = \sigma^2 \neq 0$. In fact, if $\sigma^2 = 0$, i.e. X constant, the bias would be 0, which is Proposition E.4.

Subtracting the quantities, we obtain

$$\begin{aligned}\mathbb{E}[\hat{R}(\hat{\mu})] - R(\hat{\mu}) &= -\frac{2}{n}\mathbb{E}(X^2) - (2 - \frac{2}{n})\mathbb{E}^2(X) + 2\mathbb{E}^2(X) \\ &= -\frac{2}{n}(\mathbb{E}(X^2) - \mathbb{E}^2(X)) \\ &= -2\frac{\sigma^2}{n} \neq 0\end{aligned}$$

yielding $\mathbb{E}[\hat{R}(\hat{\mu})] < R(\hat{\mu})$. □

Thus, the error of an estimator computed with the empirical risk in the same dataset is lower than the true one, i.e. saremo più generosi nel valutare i nostri estimatori. Concretamente, il risultato ci sta dicendo che se vogliamo valutare l'estimator $\hat{\mu}$ per stimare la mean (che sappiamo essere il migliore, ma immaginiamo per un attimo ce ne fossero anche altri), se lo valutiamo sullo stesso sample con cui l'abbiamo calcolato, avremo un punteggio che in media è più generoso del punteggio vero. In particolare, se ripetiamo questo esperimento molte volte cioè con molti campioni diversi della stessa distribuzione, daremo un punteggio in media $2\sigma^2/n$ più alto del dovuto. Quindi, il bias che ottengo si comporta come segue: è più alto se i miei dati tendono a variare di più, e più basso all'aumentare della size del mio dataset. Entrambi i risultati sono intuitivi: se il mio dataset varia molto, allora il problema del fittare i dati di training è più grave, perché a parità di n copriranno una varianza minore della distribuzione vera; dall'altro lato, se ho più dati copro una varianza maggiore sempre.
Considerazioni. Ora, ammettiamo che questi risultati sul bias siano estendibili ad un estimatore più generico, cioè prendiamo un certo numero di reti neurali che vogliono predire una statistica y dei miei dati, le quali hanno qualche specificazione che le differenzia (quindi sono estimatori diversi). Se io fitto le mie reti su un dataset e valuto l'estimator risultante sul dataset stesso, sarò più generosa.

Remark E.7 (Aleatorietà della rete neurale)**.** *Il rischio (o inverso del punteggio) di una rete neurale (con init e altri hyperparameters fissi) è una variabile aleatoria per la presenza del dataset, non per i parametri interni. I parametri, dato un dataset, sono fissi, una volta fissata una rete neurale e quindi la sua architettura, inizializzazione, step size, senza regolarizzazioni che danno aleatorietà (ci torna?). Quindi, quando parliamo della sua varianza qui ci riferiamo alla varianza dei dati.*

In particolare, se io valuto tantissimi estimator diversi looking for un estimator con un rischio abbastanza basso, la probabilità che io lo trovi aumenta all'aumentare del numero di estimator che valuto, ma magari lo trovo non perché il suo rischio vero è basso, ma per aleatorietà della misura di rischio.

Mi sembrava, da come parlava il prof, che questo si ricollegasse al fatto di avere tanti parametri in una rete. Come se l'aumento in numero di parametri della rete corrispondesse al mio valutare più estimatori. Perché di fatto, se aumento il numero di parametri, gli estimatori che quella rete mi può dare aumentano esponenzialmente, però i parametri sono fissi una volta dato un dataset.

Comunque, ammettendo questo parallelismo, io starei dicendo: il tuo stimatore, se lo valuti sullo stesso dataset, performa peggio di quanto credi. C'è però la possibilità di ammettere comunque questa misura di errore? Quant'è grande questo bias?

E' grande tanto quanto la varianza e piccolo tanto quanto la grandezza del tuo dataset.

Considerando la varianza come data, quello che posso fare è aumentare la misura del mio dataset. However, se stiamo facendo tutto questo per non avere bisogno del test set e ridurre i costi di fare machine learning, siamo punto a capo.